# IP on Ethernet

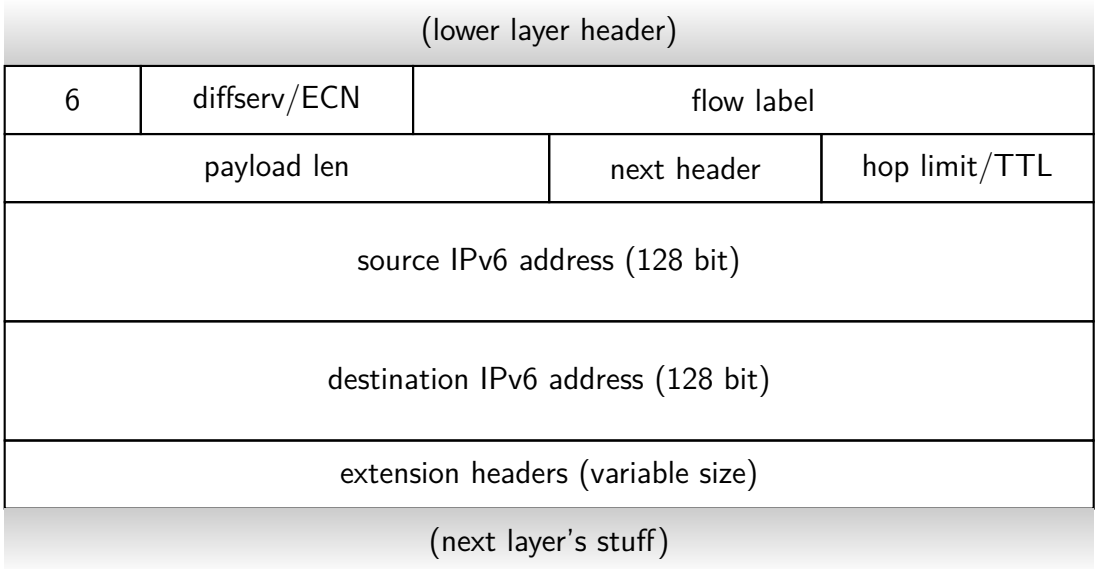
talked about how Ethernet works

but most Ethernet frames going to contain IP packets

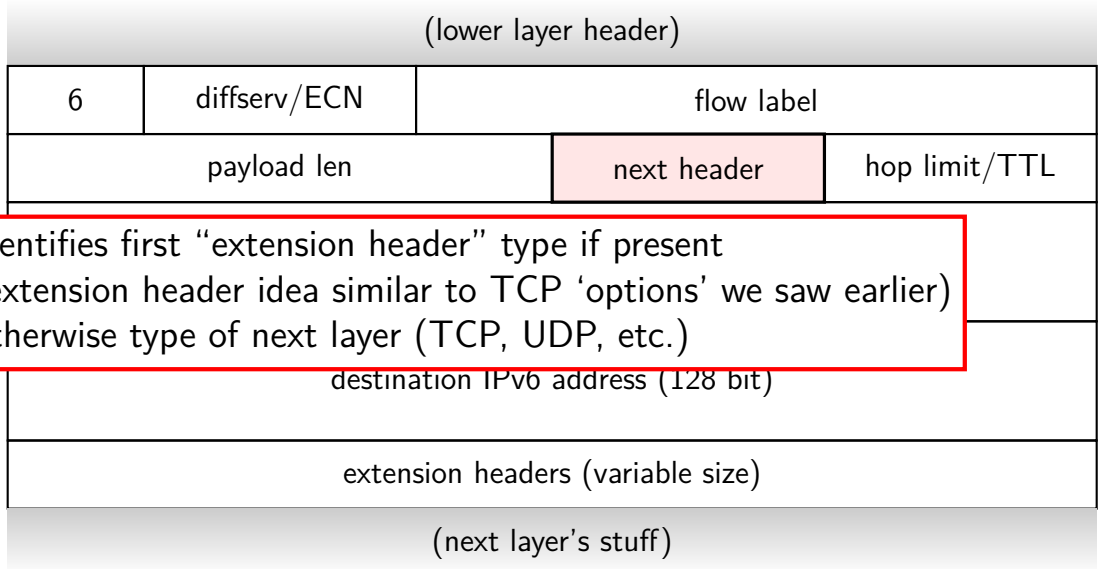
IP has its own idea of addresses

- used alongside MAC addresses

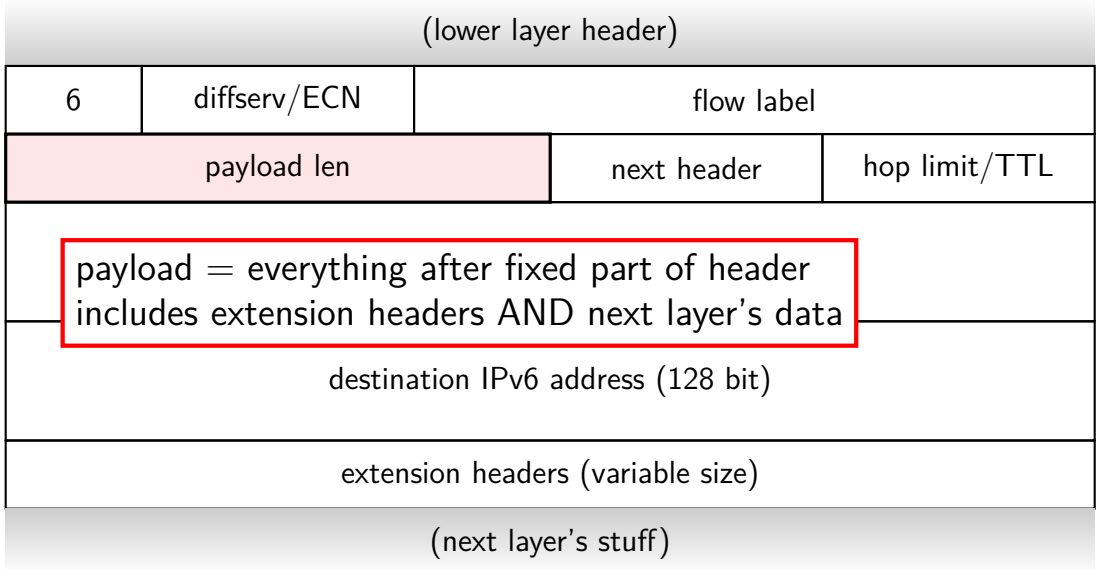
# IPv6 packet format



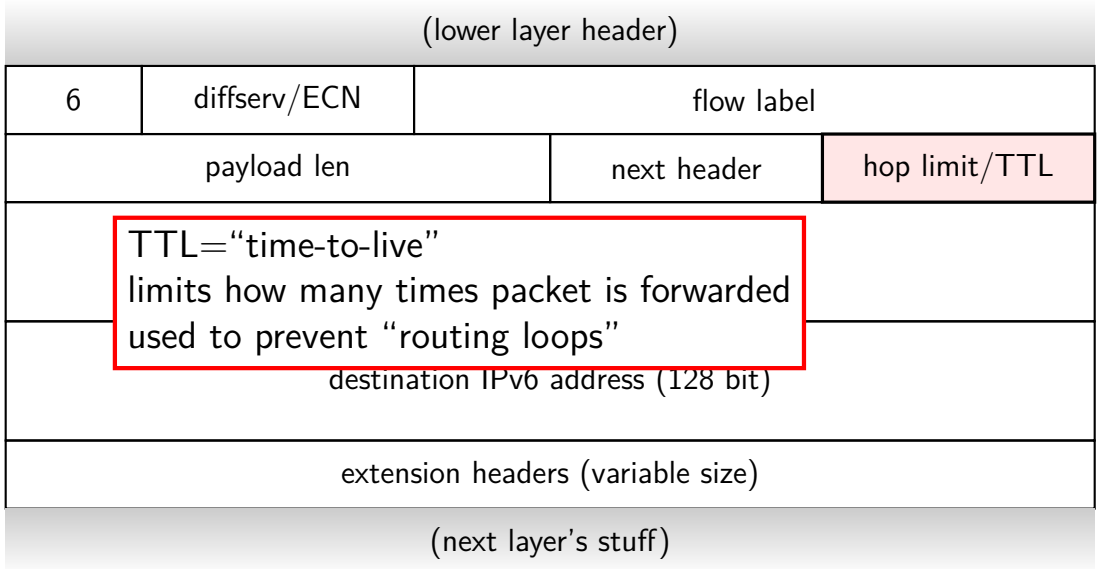
# IPv6 packet format



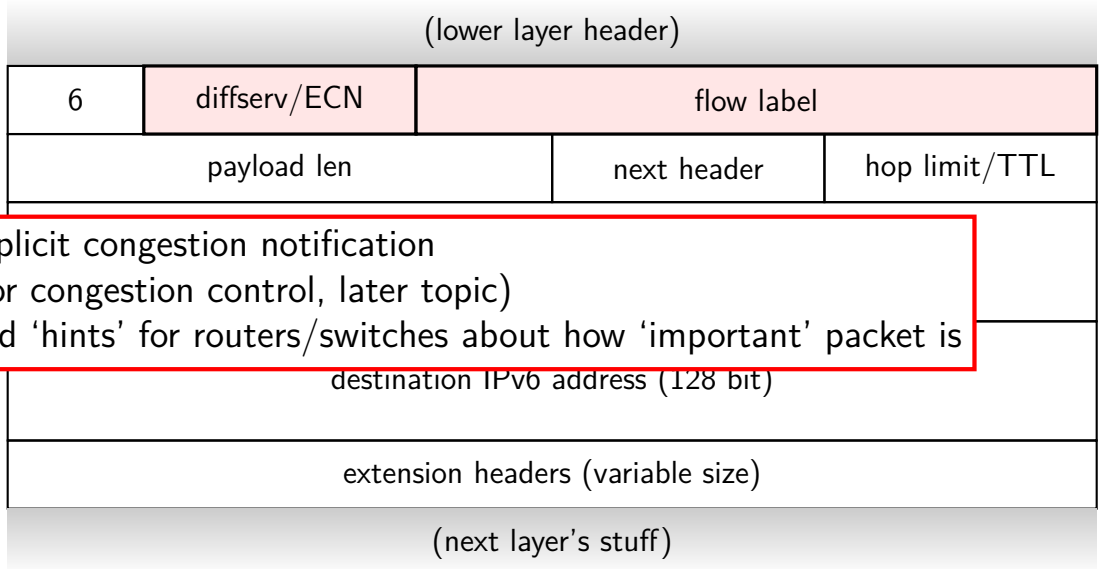
# IPv6 packet format



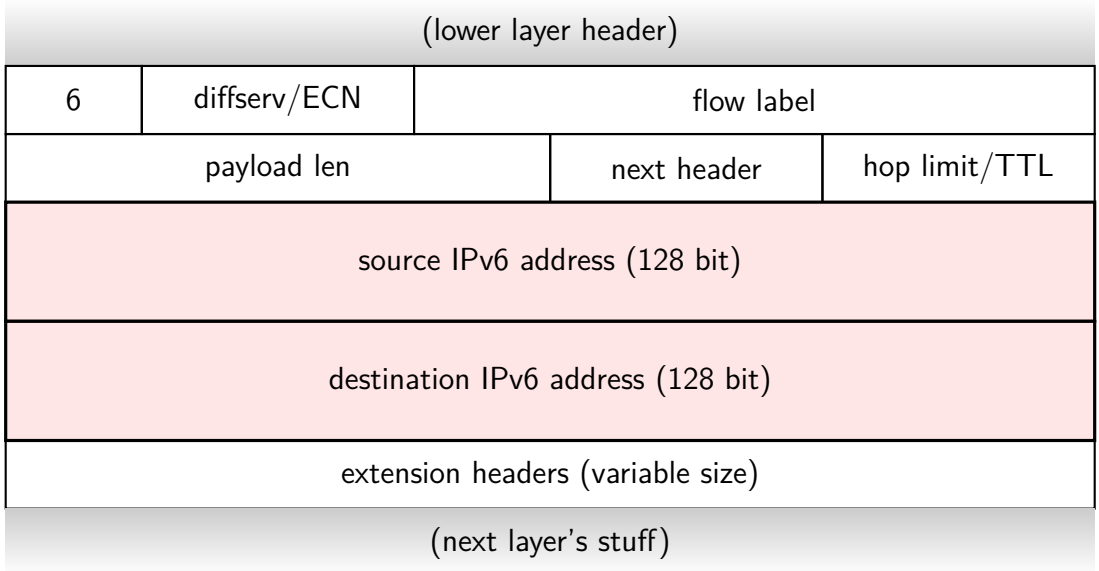
# IPv6 packet format



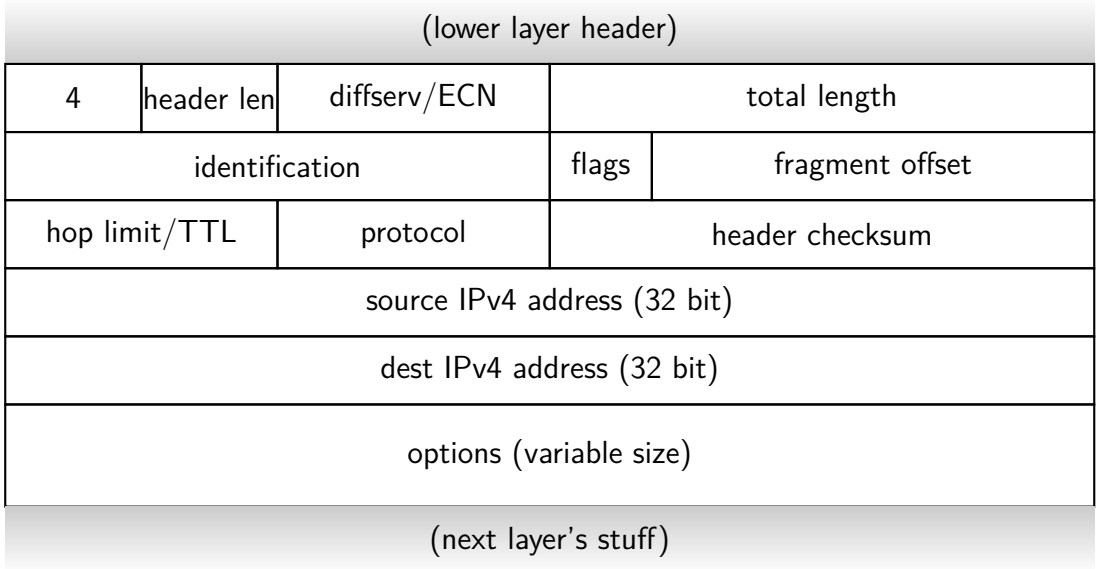
# IPv6 packet format



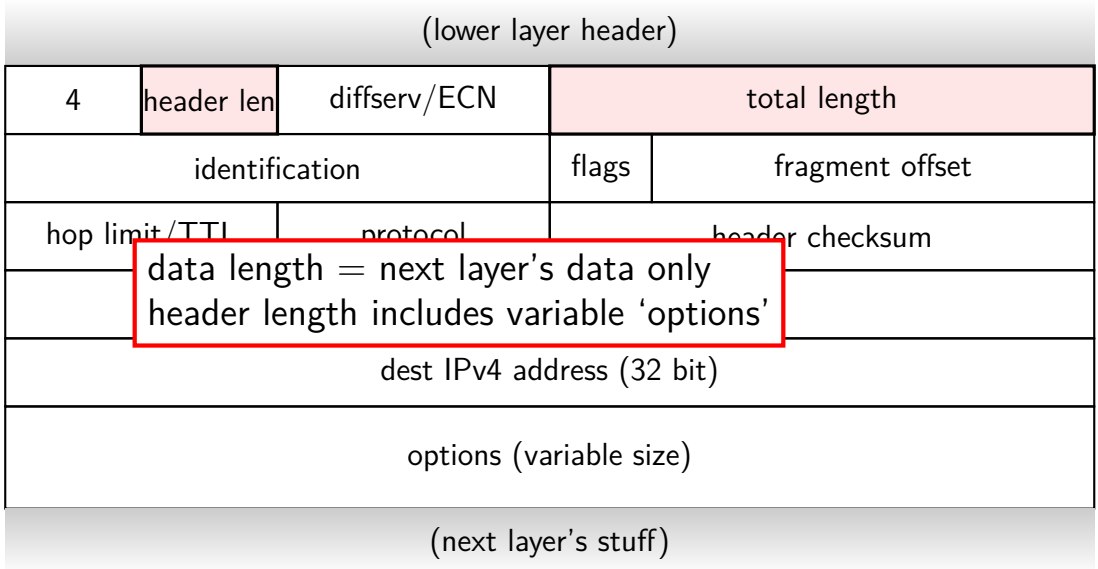
# IPv6 packet format



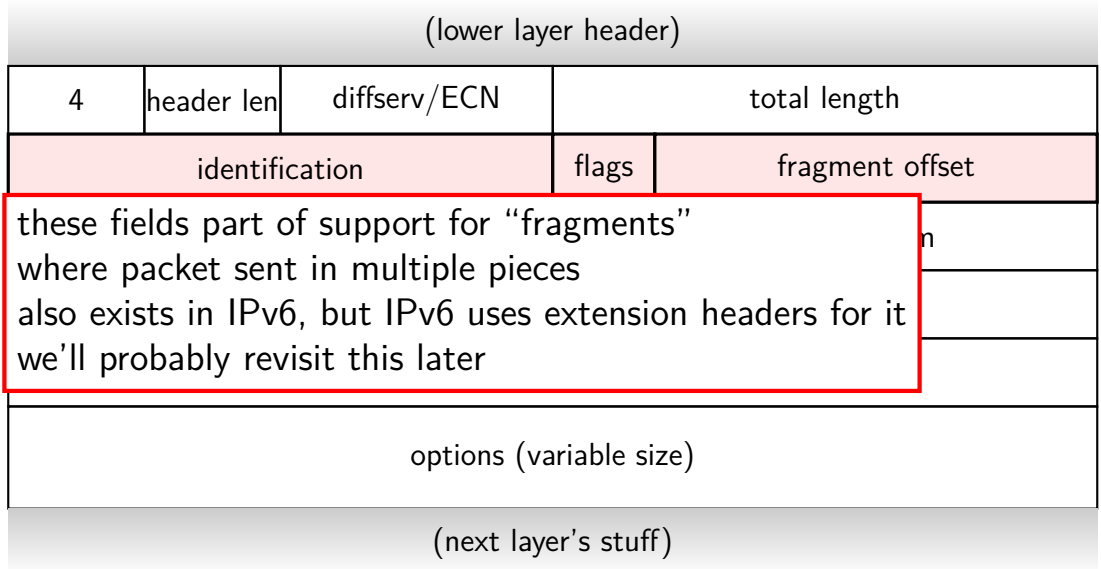
# IPv4 packet format



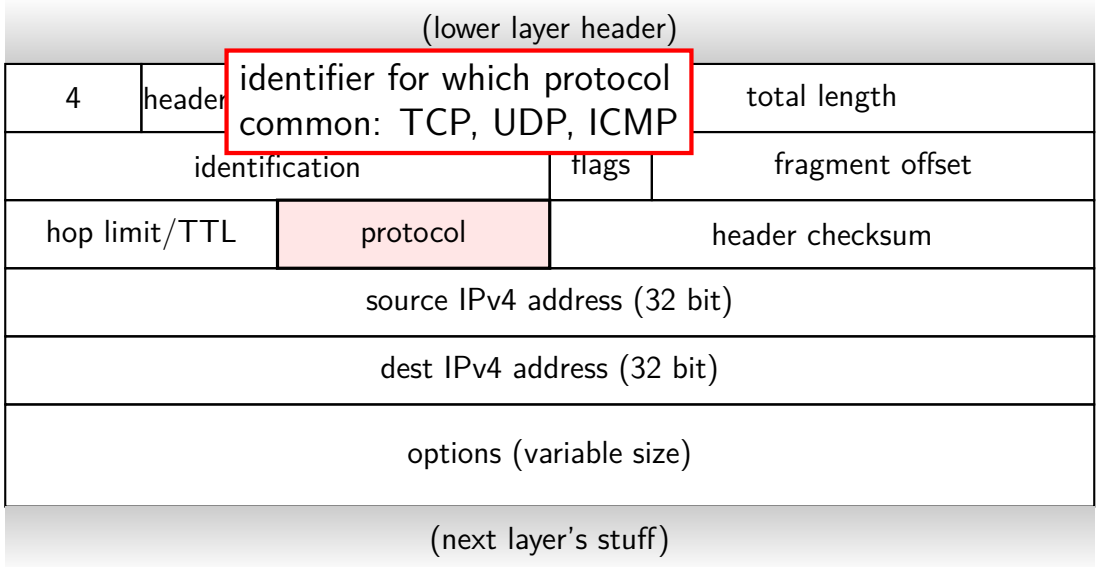
# IPv4 packet format



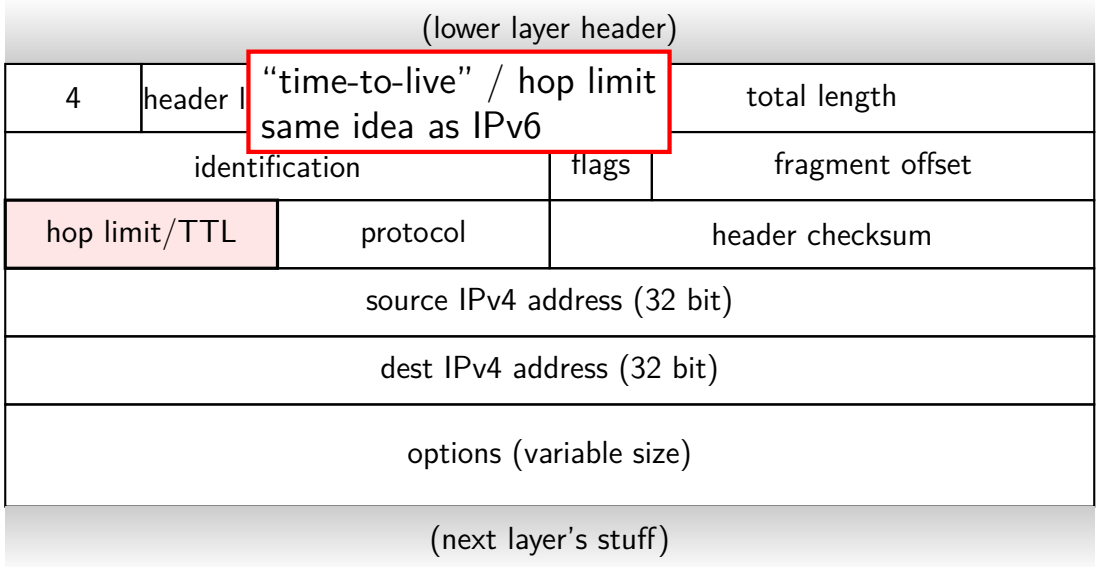
# IPv4 packet format



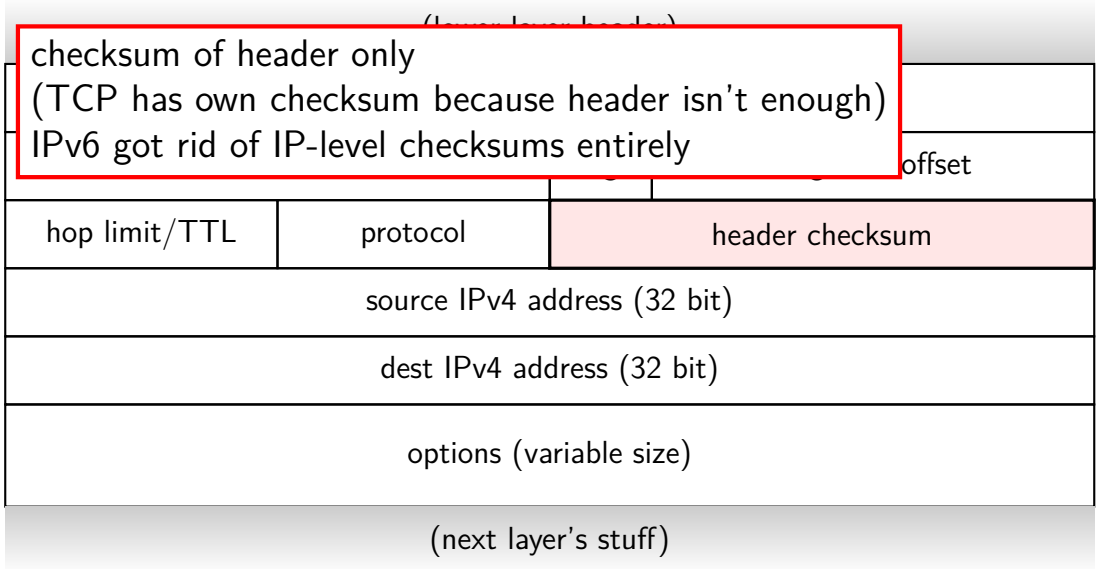
# IPv4 packet format



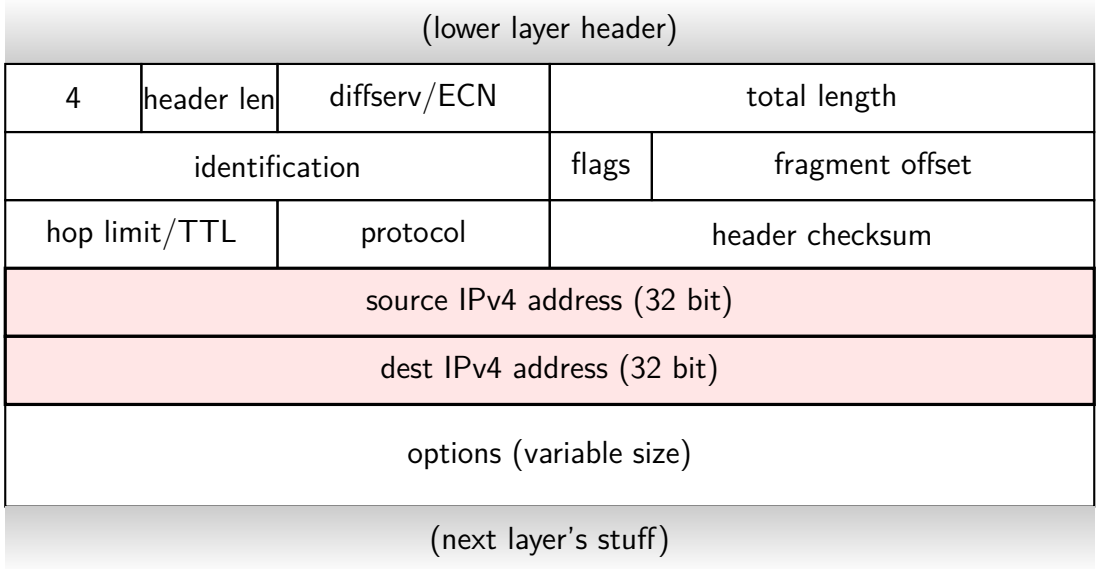
# IPv4 packet format



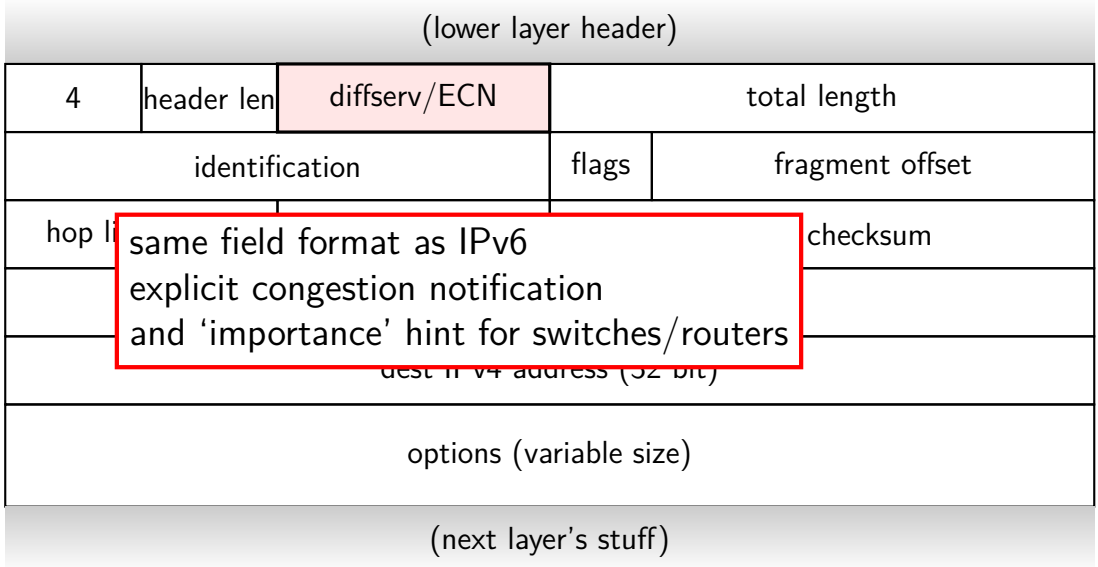
# IPv4 packet format



# IPv4 packet format



# IPv4 packet format



# IPv4 addresses

32-bit numbers

typically written like 128.143.67.11

- four 8-bit decimal values separated by dots
- first part is most significant
- same as $128 \cdot 256^3 + 143 \cdot 256^2 + 67 \cdot 256 + 11 = 2\,156\,782\,459$

# IPv4 address blocks

often will want to talk about group of IPv4 addresses

example: 128.143.67.64—128.143.67.127 (inclusive)

# IPv4 address blocks

often will want to talk about group of IPv4 addresses

example: 128.143.67.64—128.143.67.127 (inclusive)

```
10000000 10001111 01000011 00100000
10000000 10001111 01000011 00111111
```

# IPv4 address blocks

often will want to talk about group of IPv4 addresses

example: 128.143.67.64—128.143.67.127 (inclusive)

*10000000 10001111 01000011 001*00000

*10000000 10001111 01000011 001*11111

first 27 bits always same; anything for last bits

# IPv4 address blocks

often will want to talk about group of IPv4 addresses

example: 128.143.67.64—128.143.67.127 (inclusive)

`10000000 10001111 01000011 001`*`00000`*

`10000000 10001111 01000011 001`*`11111`*

first 27 bits always same; anything for last bits

# IPv4 address blocks

often will want to talk about group of IPv4 addresses

example: 128.143.67.64—128.143.67.127 (inclusive)

`10000000 10001111 01000011 00100000`

`10000000 10001111 01000011 00111111`

first 27 bits always same; anything for last bits

more convenient representation: 128.143.67.64/27

called "CIDR notation"

CIDR = classless inter-domain routing (will come up when we discuss routing)

# CIDR notation examples

5.7.3.3/14 = 5.4.0.0/14 = 5.4.0.0—5.7.255.255

128.143.0.0/16 = 128.143.0.0—128.143.255.255

192.168.0.0/24 = 192.168.0.0—192.168.0.255

10.0.0.0/8 = 10.0.0.0–10.255.255.255

# CIDR notation examples

5.7.3.3/14 = 5.4.0.0/14 = 5.4.0.0—5.7.255.255

*also written 5.4/14*

128.143.0.0/16 = 128.143.0.0—128.143.255.255

*also written 128.143/16*

192.168.0.0/24 = 192.168.0.0—192.168.0.255

10.0.0.0/8 = 10.0.0.0–10.255.255.255

*also written 10/8*

# alternate notation: netmasks

instead of writing 128.143.67.64/27 might say

128.143.67.64 and "network mask" of 255.255.255.224

255.255.255.224 = 27 1's

# alternate notation: netmasks

instead of writing 128.143.67.64/27 might say

128.143.67.64 and "network mask" of 255.255.255.224

255.255.255.224 = 27 1's

if some-address bitwise-AND netmask = 128.143.67.64
bitwise-AND netmask,
then some-address is in the range

# IPv6 addresses

IPv6 like IPv4, but with 128-bit numbers

written in hex, 16-bit parts, seperated by colons (`:`)

strings of 0s represented by double-colons (`::`)

typically given to users in blocks of $2^{80}$ or $2^{64}$ addresses

`2607:f8b0:400d:c00::6a` =
`2607:f8b0:400d:0c00:0000:0000:0000:006a`

$$\texttt{2607f8b0400d0c0000000000000006a}_{\text{SIXTEEN}}$$

# IPv6 CIDR notation examples

2607:fb80:400d:0c00::/64 =

`2607:fb80:400d:0c00:0000:0000:0000:0000`—
`2607:fb80:400d:0c00:ffff:ffff:ffff:ffff`

2607:fb80::/30 =

`2607:fb80:0000:0000:0000:0000:0000:0000`—
`2607:fb83:ffff:ffff:ffff:ffff:ffff:ffff`

Internet Assigned Numbers Authority

# IANA IPv4 Address Space Registry

**Last Updated**

2023-12-18

**Registration Procedure(s)**

Allocations to RIRs are made in line with the Global Policy published at [http://www.icann.org/en/resources/policy/global-addressing]. All other assignments require IETF Review.

**Description**

The allocation of Internet Protocol version 4 (IPv4) address space to various registries is listed here. Originally, all the IPv4 address spaces was managed directly by the IANA. Later parts of the address space were allocated to various other registries to manage for particular purposes or regional areas of the world. RFC 1466 [RFC1466] documents most of these allocations.

**Reference**

[RFC7249]

**Available Formats**

CSV XML HTML Plain text

| Prefix | Designation | Date | WHOIS | RDAP | Status [1] | Note |
|---|---|---|---|---|---|---|
| 000/8 | IANA - Local Identification | 1981-09 | | | RESERVED | [2][3] |
| 001/8 | APNIC | 2010-01 | whois.apnic.net | https://rdap.apnic.net/ | ALLOCATED | |
| 002/8 | RIPE NCC | 2009-09 | whois.ripe.net | https://rdap.db.ripe.net/ | ALLOCATED | |
| 003/8 | Administered by ARIN | 1994-05 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |

| Prefix | Designation | Date | WHOIS | RDAP | Status [1] | Note |
|---|---|---|---|---|---|---|
| 000/8 | IANA - Local Identification | 1981-09 | | | RESERVED | [2][3] |
| 001/8 | APNIC | 2010-01 | whois.apnic.net | https://rdap.apnic.net/ | ALLOCATED | |
| 002/8 | RIPE NCC | 2009-09 | whois.ripe.net | https://rdap.db.ripe.net/ | ALLOCATED | |
| 003/8 | Administered by ARIN | 1994-05 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |
| 004/8 | Administered by ARIN | 1992-12 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |
| 005/8 | RIPE NCC | 2010-11 | whois.ripe.net | https://rdap.db.ripe.net/ | ALLOCATED | |
| 006/8 | Army Information Systems Center | 1994-02 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |
| 007/8 | Administered by ARIN | 1995-04 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |
| 008/8 | Administered by ARIN | 1992-12 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |
| 009/8 | Administered by ARIN | 1992-08 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |
| 010/8 | IANA - Private Use | 1995-06 | | | RESERVED | [4] |
| 011/8 | DoD Intel Information Systems | 1993-05 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |
| 012/8 | AT&T Bell Laboratories | 1995-06 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |
| 013/8 | Administered by ARIN | 1991-09 | whois.arin.net | https://rdap.arin.net/registry http://rdap.arin.net/registry | LEGACY | |

(and 241 more)

Internet Assigned Numbers Authority

# Internet Protocol Version 6 Address Space

**Last Updated**
2019-09-13

**Note**
The IPv6 address management function was formally delegated to IANA in December 1995 [RFC1881]. The registration procedure was confirmed with the IETF Chair in March 2010.

| IPv6 Prefix | Allocation | Reference | Notes |
|---|---|---|---|
| 0000::/8 | Reserved by IETF | [RFC3513] [RFC4291] | [1] [2] [3] [4] [5] [6] |
| 0100::/8 | Reserved by IETF | [RFC3513] [RFC4291] | 0100::/64 reserved for Discard-Only Address Block [RFC6666]. Complete |
| 0200::/7 | Reserved by IETF | [RFC4048] | Deprecated as of December 2004 [RFC4048]. Formerly an OSI NSAP-map |
| 0400::/6 | Reserved by IETF | [RFC3513] [RFC4291] | |
| 0800::/5 | Reserved by IETF | [RFC3513] [RFC4291] | |
| 1000::/4 | Reserved by IETF | [RFC3513] [RFC4291] | |
| 2000::/3 | Global Unicast | [RFC3513] [RFC4291] | The IPv6 Unicast space encompasses the entire IPv6 address range with are currently limited to the IPv6 unicast address range of 2000::/3. IANA *address-assignments*]. [7] [8] [9] [10] [11] [12] [13] [14] [15] |
| 4000::/3 | Reserved by IETF | [RFC3513] [RFC4291] | |

# IPv6 Global Unicast Address Assignments

**Last Updated**
2024-07-23

**Registration Procedure(s)**
Allocations to RIRs are made in line with the Global Policy published at [http://www.icann.org/en/resources/policy/global-addressing].
All other assignments require IETF Review.

**Description**
The allocation of Internet Protocol version 6 (IPv6) unicast address space is here. References to the various other registries detailing the use of the IPv space can be found in the [IPv6 Address Space registry].

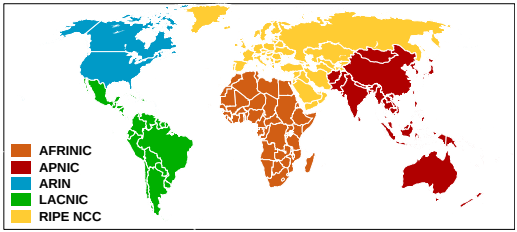
**Reference**
[RFC7249]

**Note**
The assignable Global Unicast Address space is defined in [RFC3513] as the ad defined by the prefix 2000::/3. [RFC3513] was later obsoleted by [RFC4291]. A space in this block not listed in the table below is reserved by IANA for fut allocation.

**Available Formats**
CSV XML HTML Plain text

| Prefix | Designation | Date | WHOIS | RDAP |
|---|---|---|---|---|
| 2001:0000::/23 | IANA | 1999-07-01 | whois.iana.org | |
| 2001:0200::/23 | APNIC | 1999-07-01 | whois.apnic.net | https://rdap.apnic.net/ |
| 2001:0400::/23 | ARIN | 1999-07-01 | whois.arin.net | https://rdap.arin.net/regist http://rdap.arin.net/rdap |
| 2001:0600::/23 | RIPE NCC | 1999-07-01 | whois.ripe.net | https://rdap.db.ripe.net/ |

# regional internet registries (RIRs)

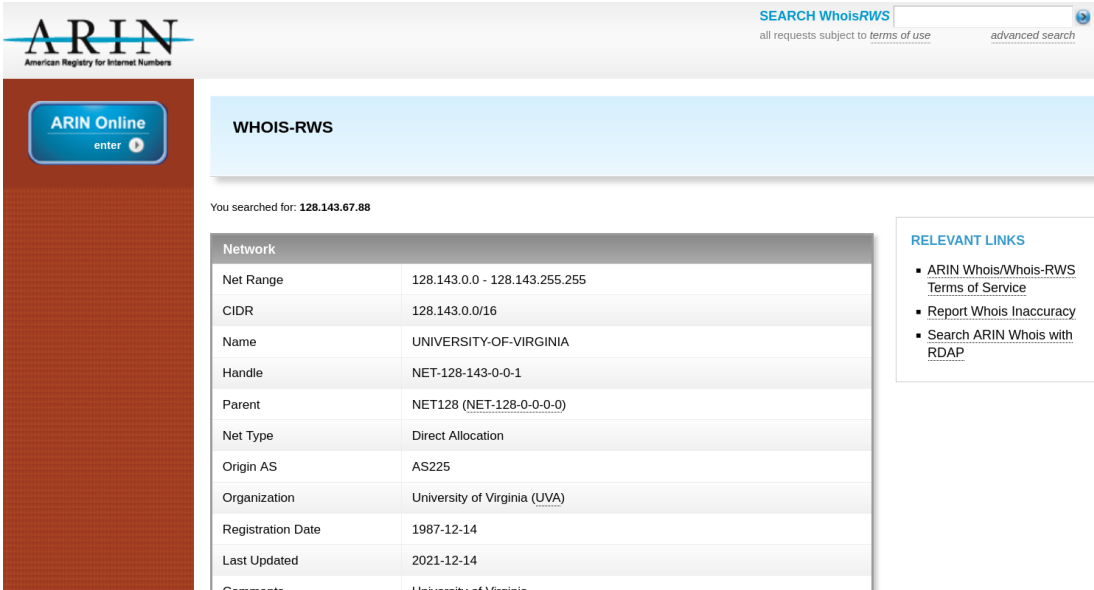



map from Wikimedia Commons,
users Dork, Canuckguy et al, Sémhur, CC-BY-SA 3.0

most useful addresses managed by RIRs

- African Network Information Centre (AFRINIC)
- American Registry for Internet Numbers (ARIN)
- Asia Pacific Network Information Centre (APNIC)
- Latin American and Carribean Network Information Centre (LACNIC)
- Réseaux IP Européens Network Coordination Centre (RIPE NCC)

# RIR suballocations



# special IPv4 addresses




## IANA IPv4 Special-Purpose Address Registry

**Created**
2009-08-19
**Last Updated**
2021-02-04
**Available Formats**

# special IPv4 addresses

| Address Block | Name | RFC | Allocation Date | Termination Date | Source | Destination | For |
|---|---|---|---|---|---|---|---|
| 0.0.0.0/8 | "This network" | [RFC791], Section 3.2 | 1981-09 | N/A | True | False | Fals |
| 0.0.0.0/32 | "This host on this network" | [RFC1122], Section 3.2.1.3 | 1981-09 | N/A | True | False | Fals |
| 10.0.0.0/8 | Private-Use | [RFC1918] | 1996-02 | N/A | True | True | True |
| 100.64.0.0/10 | Shared Address Space | [RFC6598] | 2012-04 | N/A | True | True | True |
| 127.0.0.0/8 | Loopback | [RFC1122], Section 3.2.1.3 | 1981-09 | N/A | False [1] | False [1] | Fals |
| 169.254.0.0/16 | Link Local | [RFC3927] | 2005-05 | N/A | True | True | Fals |
| 172.16.0.0/12 | Private-Use | [RFC1918] | 1996-02 | N/A | True | True | True |
| 192.0.0.0/24 [2] | IETF Protocol Assignments | [RFC6890], Section 2.1 | 2010-01 | N/A | False | False | Fals |
| 192.0.0.0/29 | IPv4 Service Continuity Prefix | [RFC7335] | 2011-06 | N/A | True | True | True |
| 192.0.0.8/32 | IPv4 dummy address | [RFC7600] | 2015-03 | N/A | True | False | Fals |
| 192.0.0.9/32 | Port Control Protocol Anycast | [RFC7723] | 2015-10 | N/A | True | True | True |
| 192.0.0.10/32 | Traversal Using Relays around NAT Anycast | [RFC8155] | 2017-02 | N/A | True | True | True |
| 192.0.0.170/32, 192.0.0.171/32 | NAT64/DNS64 Discovery | [RFC8880] [RFC7050], Section 2.2 | 2013-02 | N/A | False | False | Fals |
| 192.0.2.0/24 | Documentation (TEST- | [RFC5737] | 2010-01 | N/A | False | False | Fals |

# selected special IP addresses

loopback (current machine) — `127/8` (v4), `::1/128` (v6)

link-local (current network only) —
`169.254/16` (v4), `ff80::/10` (v6)

private use (non-public networks only) —
`192.168/16`, `172.16/12`, `10/8` (v4), (kinda) `fc00::/7` (v6)

multicast groups and related — `224/4` (v4), `ff00::/8` (v6)
multiple nodes can be part of a single "multicast group"

broadcast (all on current network) —
`255.255.255.255`, `ff01::1`

"future use" —
rest of `240/4` (v4), `4000::`—`efff::` (v6)

# which link local?

"link local": `169.254/16`, `fe80::/10`

specific to each local network

`fe80::17` on network A != `fe80::17` on network B

problem: machine can be connected to two networks

# which link local?

"link local": `169.254/16`, `fe80::/10`

specific to each local network

`fe80::17` on network A != `fe80::17` on network B

problem: machine can be connected to two networks

solution: `fe80::17%A` versus `fe80::17%B`

# which link local?

"link local": `169.254/16`, `fe80::/10`

specific to each local network

`fe80::17` on network A != `fe80::17` on network B

problem: machine can be connected to two networks

solution: `fe80::17%A` versus `fe80::17%B`

what about IPv4? uh... too bad?

> "There is no standard or obvious solution to this problem...must be done explicitly through other means. The specification does not stipulate those means." — RFC 3927, section 3.2

# switch v router: tables

switch ('bridge') table

| MAC address | port |
|---|---|
| `00:11:22:33:44:55` | 1 |
| `00:33:00:01:02:aa` | 2 |
| `00:44:00:01:02:bb` | 3 |
| … | … |
| default | `(all)` |

routing table

| IP addresses | gateway | iface |
|---|---|---|
| `2001:0db8:40::/48` | `---` | `int` |
| `3fff:1000:19::/48` | `---` | `ext` |
| … | … | … |
| default | `fe80::17` | `ext` |

# switch v router: tables

switch ('bridge') table

| MAC address | port |
|---|---|
| 00:11:22:33:44:55 | 1 |
| 00:33:00:01:02:aa | 2 |
| 00:44:00:01:02:bb | 3 |
| … | … |
| default | (all) |

routing table

| IP addresses | gateway | iface |
|---|---|---|
| 2001:0db8:40::/48 | --- | int |
| 3fff:1000:19::/48 | --- | ext |
| … | … | … |
| default | fe80::17 | ext |

one logical device with multiple ports

not in table: always broadcast

# switch v router: tables

switch ('bridge') table

| MAC address | port |
|---|---|
| 00:11:22:33:44:55 | 1 |
| 00:33:00:01:02:aa | 2 |
| 00:44:00:01:02:bb | 3 |
| … | … |
| default | (all) |

routing table

| IP addresses | gateway | iface |
|---|---|---|
| 2001:0db8:40::/48 | --- | int |
| 3fff:1000:19::/48 | --- | ext |
| … | … | … |
| default | fe80::17 | ext |

'interface' = which network

one interface might have multiple ports that are 'bridged' together

# switch v router: tables

switch ('bridge') table

| MAC address | port |
|---|---|
| 00:11:22:33:44:55 | 1 |
| 00:33:00:01:02:aa | 2 |
| 00:44:00:01:02:bb | 3 |
| … | … |
| default | (all) |

routing table

| IP addresses | gateway | iface |
|---|---|---|
| 2001:0db8:40::/48 | --- | int |
| 3fff:1000:19::/48 | --- | ext |
| … | … | … |
| default | fe80::17 | ext |

gateway = who to send to next
no gateway = 'direct' to destination

need to have specific destination
to send to on interface

# trivial tables

let's say we're connected to ONE interface with ONE port

# trivial tables

let's say we're connected to ONE interface with ONE port

tables are really trivial:

switch ('bridge') table

| MAC address | port |
|---|---|
| default | the port |

routing table (IPv6)

| IP addresses | gateway | iface |
|---|---|---|
| `2001:0db8:40::/48` | `---` | `the interface` |
| default | `fe80::17` | `the interface` |

routing table (IPv4)

| IP addresses | gateway | iface |
|---|---|---|
| `192.0.2.0/24` | `---` | `the interface` |
| default | `192.0.2.1` | `the interface` |

# trivial tables

let's say we're connected to ONE interface with ONE port

tables are really trivial:

switch ('bridge') table

| MAC address | port |
|---|---|
| default | the port |

routing table (IPv6)

| IP addresses | gateway | iface |
|---|---|---|
| `2001:0db8:40::/48` | `---` | `the interface` |
| default | `fe80::17` | `the interface` |

routing table (IPv4)

| IP addresses | gateway | iface |
|---|---|---|
| `192.0.2.0/24` | `---` | `the interface` |
| default | `192.0.2.1` | `the interface` |

# trivial tables

let's say we're connected to ONE interface with ONE port

tables are really trivial:

switch ('bridge') table

| MAC address | port |
|---|---|
| default | the port |

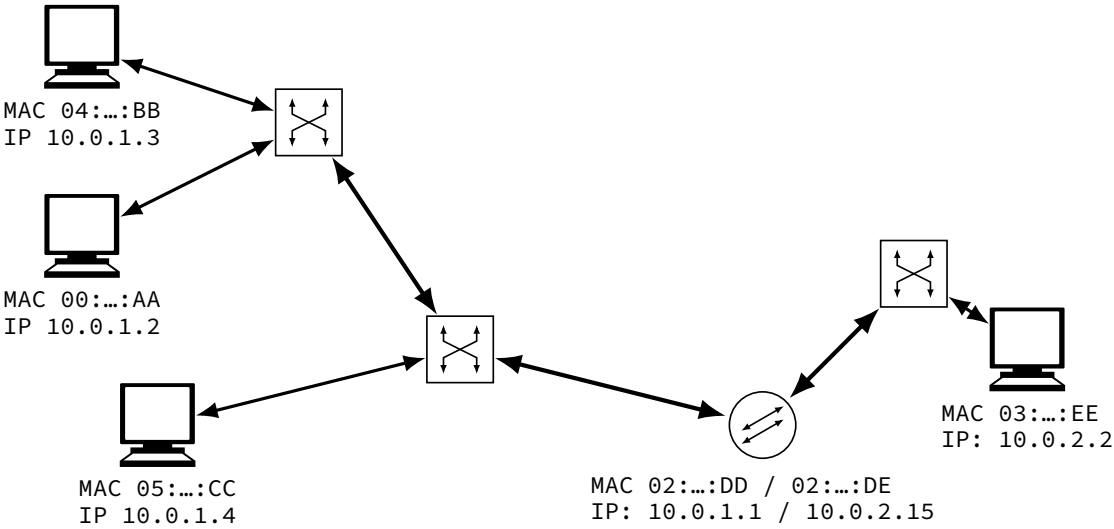
routing table (IPv6)

| IP addresses | gateway | iface |
|---|---|---|
| `2001:0db8:40::/48` | `---` | `the interface` |
| default | `fe80::17` | `the interface` |

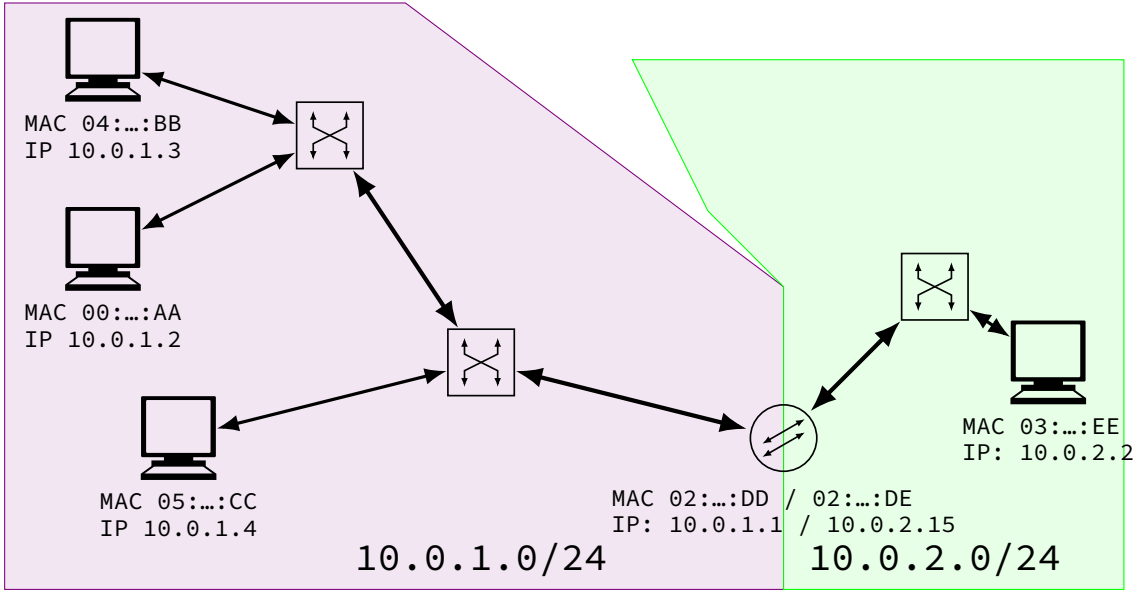
routing table (IPv4)

| IP addresses | gateway | iface |
|---|---|---|
| `192.0.2.0/24` | `---` | `the interface` |
| default | `192.0.2.1` | `the interface` |

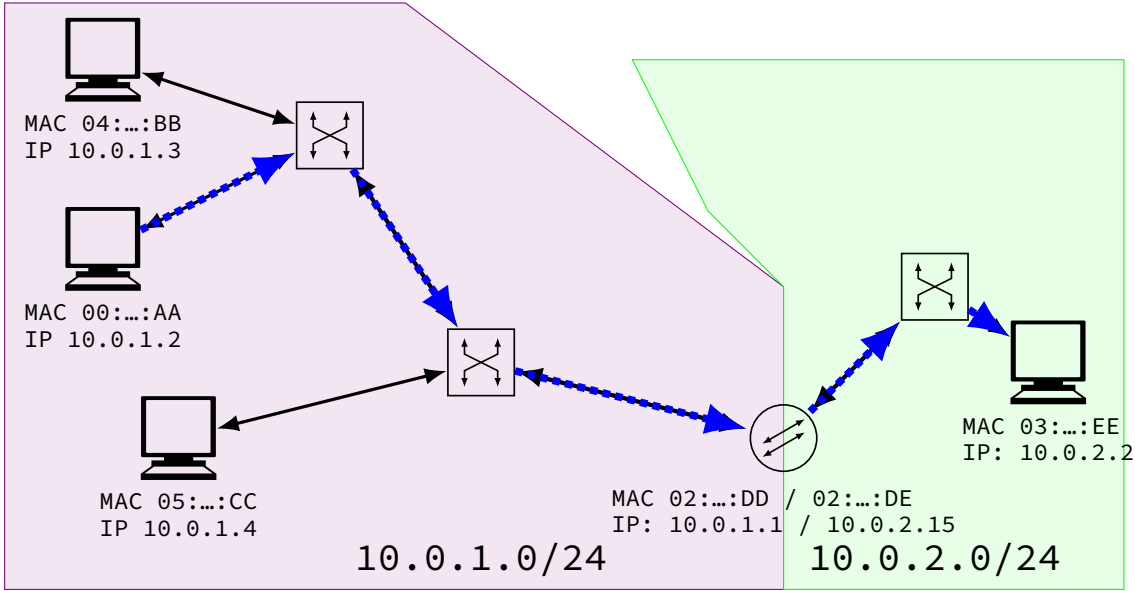
# switch v router: on the wires



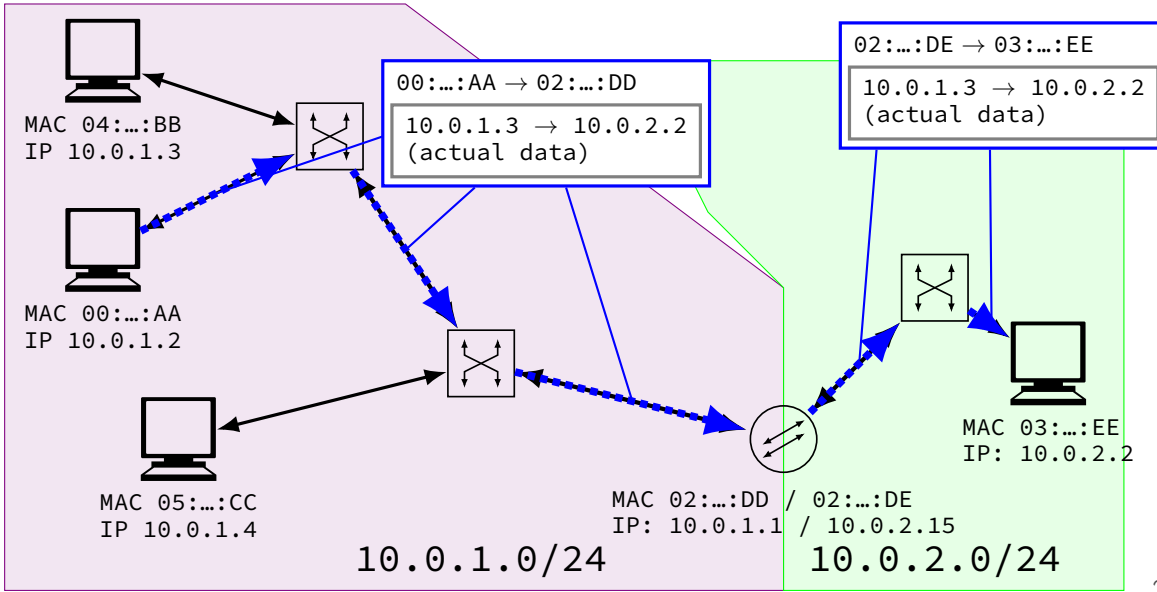
# switch v router: on the wires



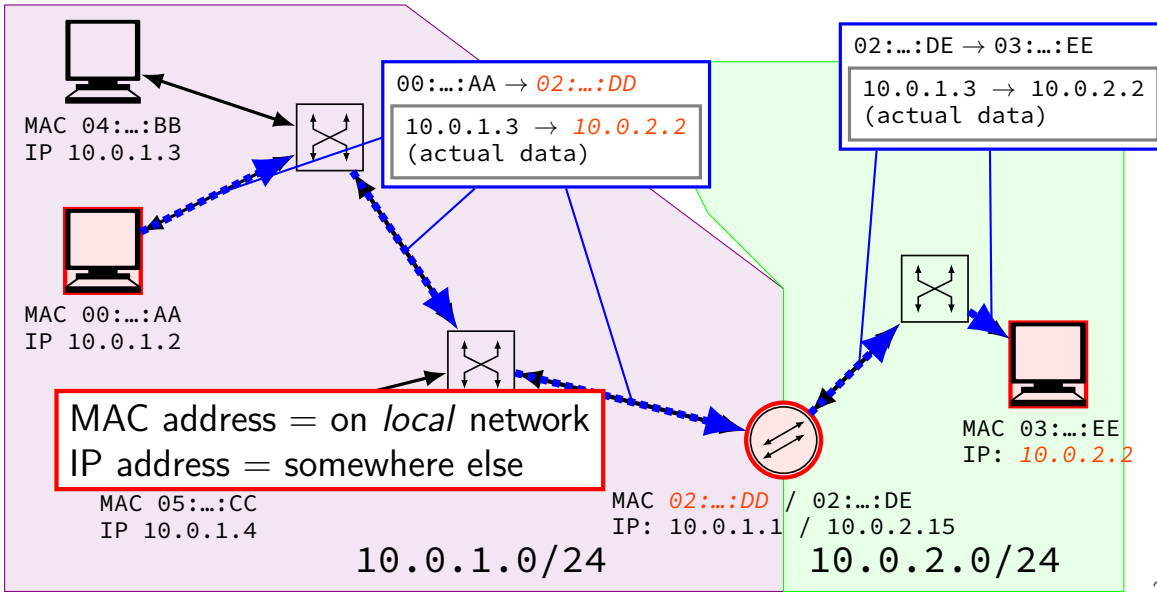
# switch v router: on the wires



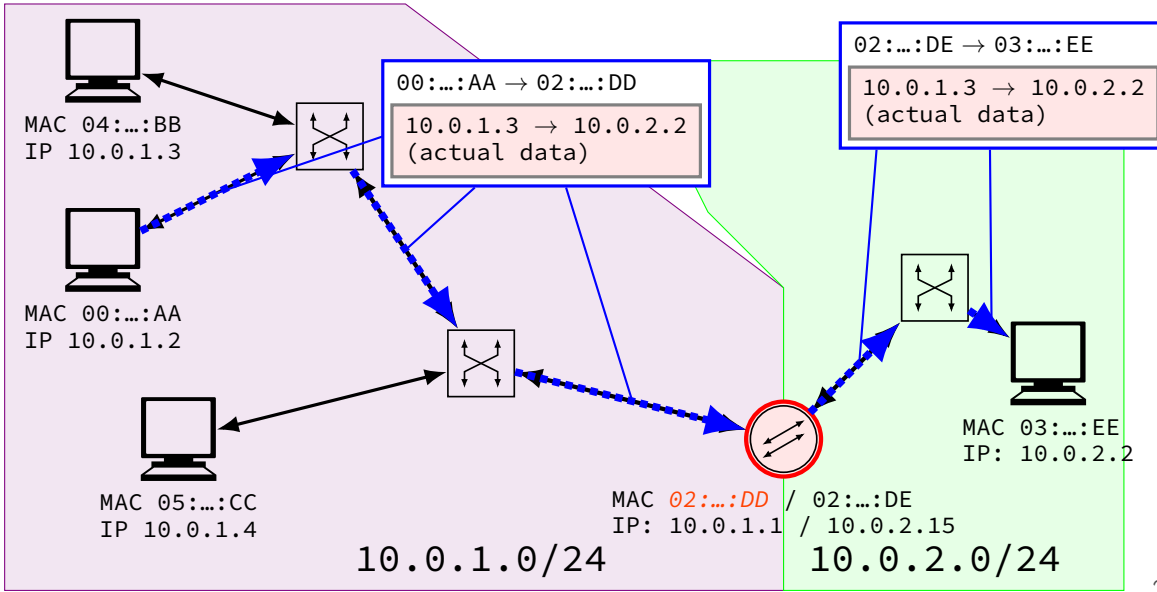
# switch v router: on the wires



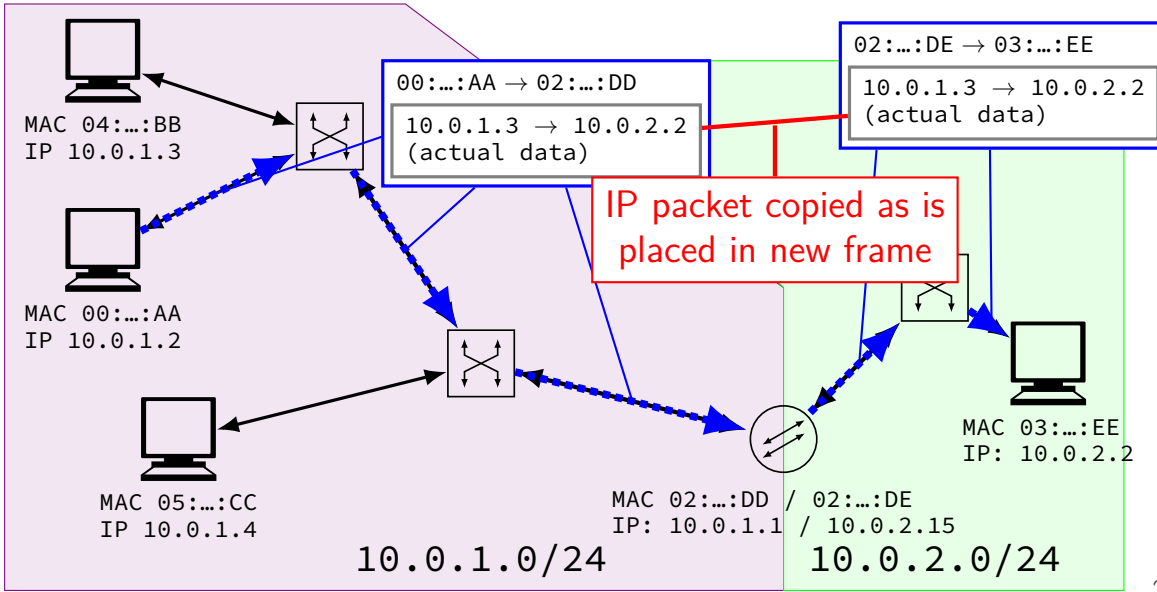
# switch v router: on the wires



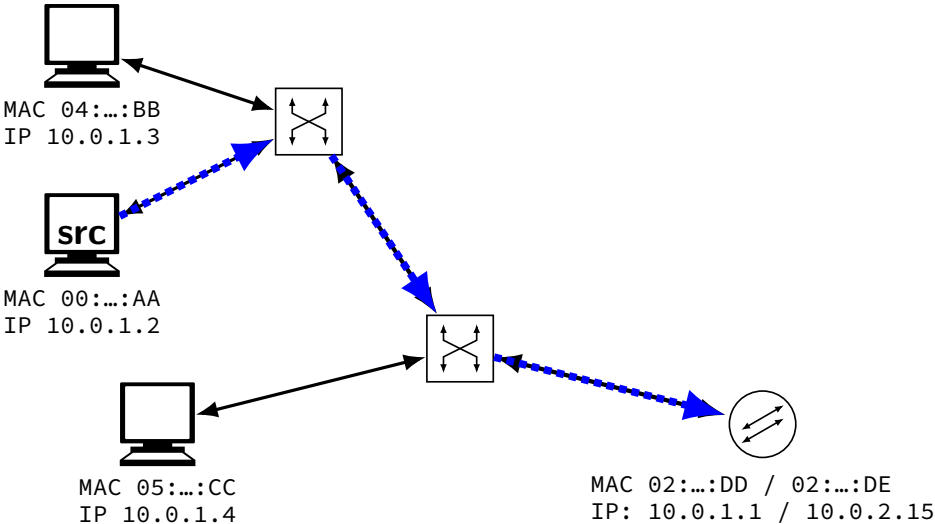
# switch v router: on the wires



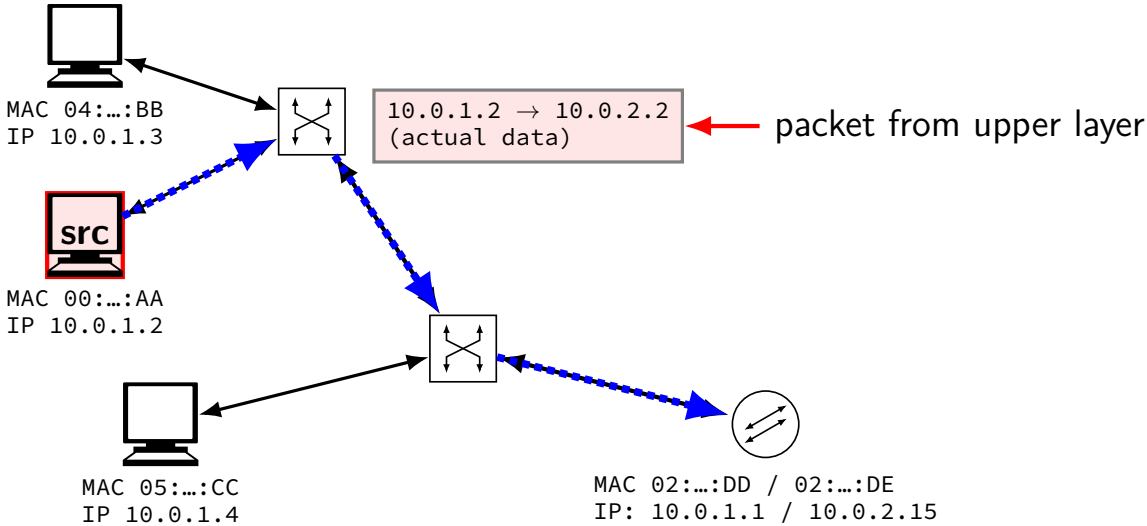
# switch v router: on the wires



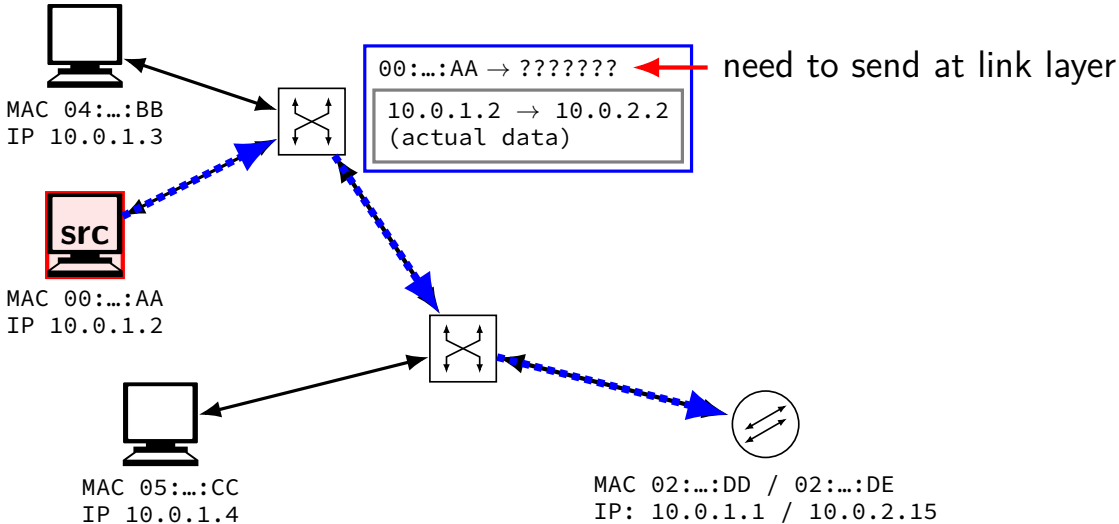
# steps at the sender



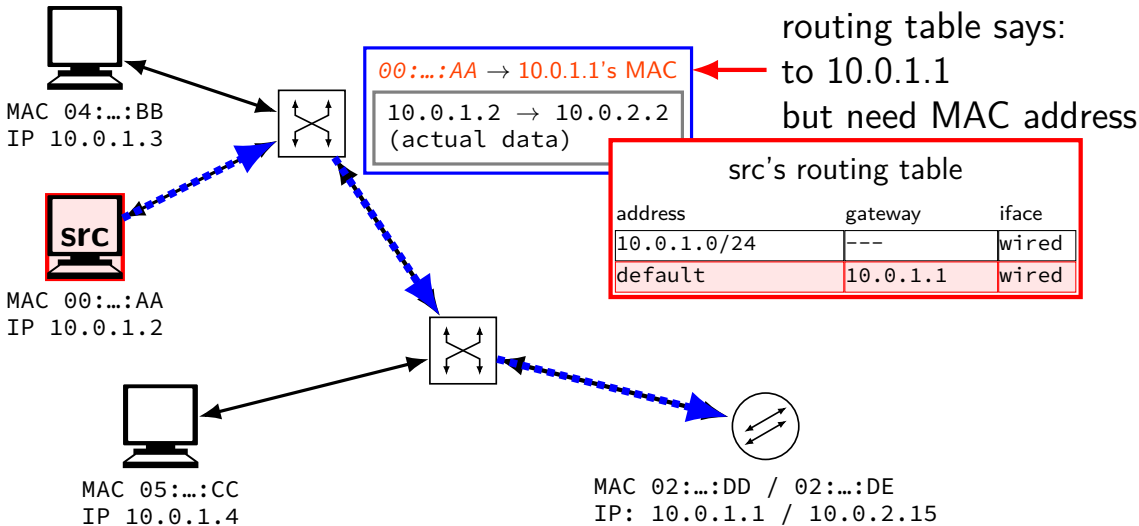
# steps at the sender



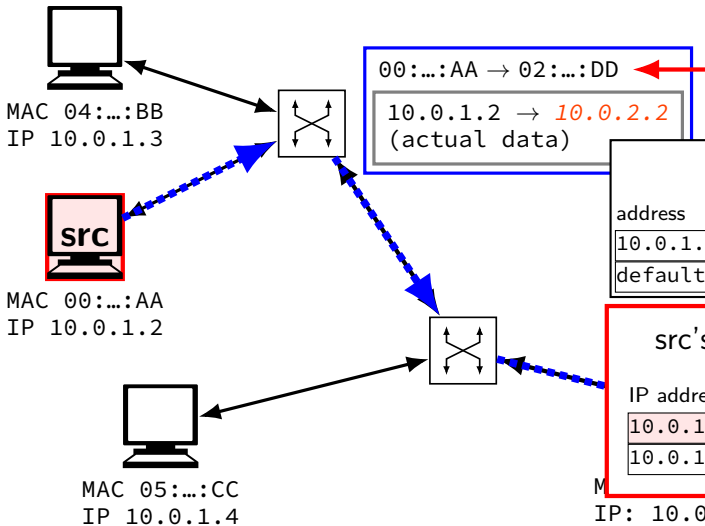
# steps at the sender



# steps at the sender



# steps at the sender




need IP:MAC address table called neighbor table or ARP table

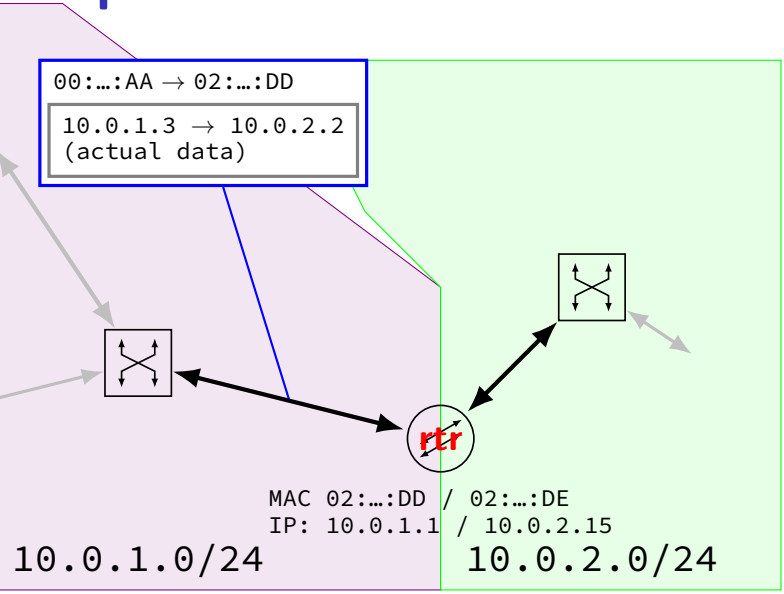
src's routing table

| address | gateway | iface |
|---|---|---|
| 10.0.1.0/24 | --- | wired |
| default | 10.0.1.1 | wired |

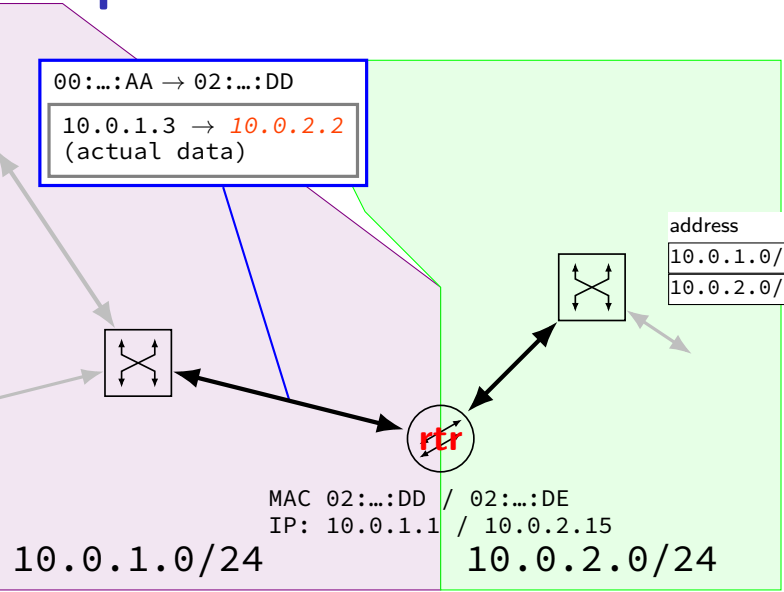
src's neighbor table

| IP address | MAC addresss |
|---|---|
| 10.0.1.1 | 05:…:DD |
| 10.0.1.4 | 05:…:CC |

# steps at the router



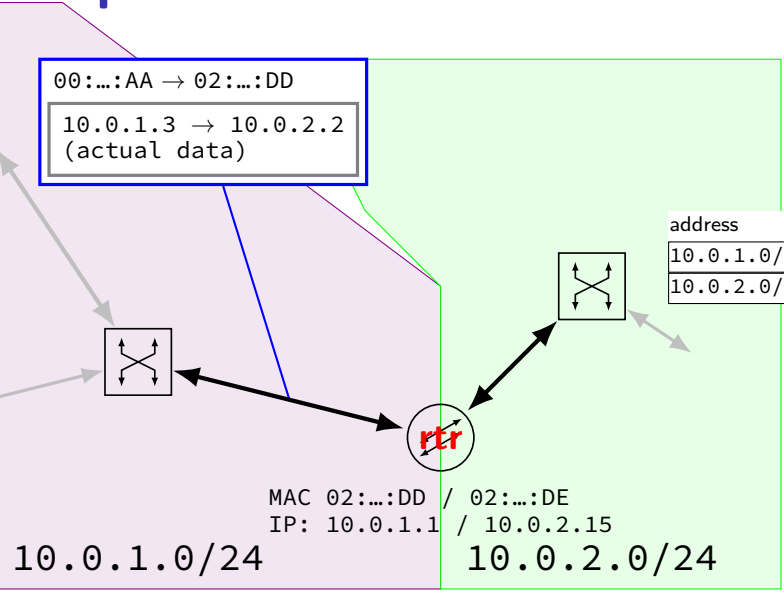
# steps at the router




rtr's routing table

| address | gateway | iface |
|---|---|---|
| 10.0.1.0/24 | --- | left |
| 10.0.2.0/24 | --- | right |

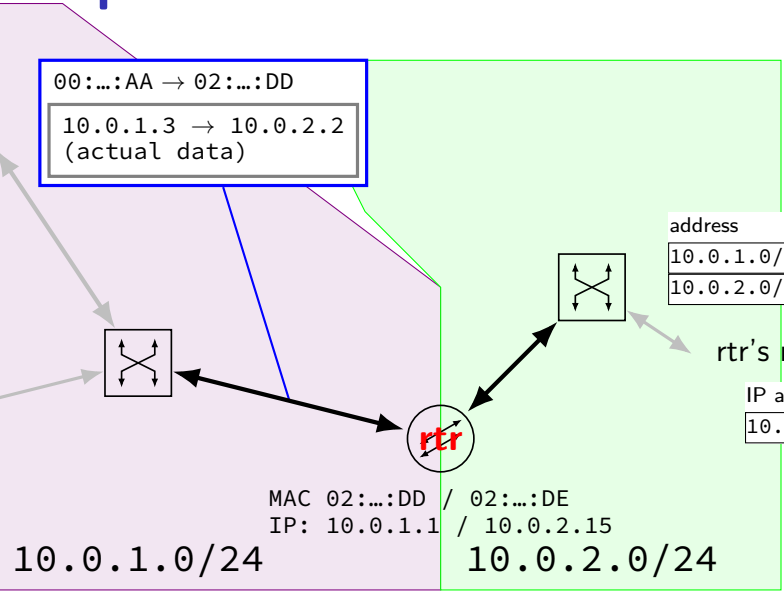
# steps at the router




rtr's routing table

| address | gateway | iface |
|---|---|---|
| 10.0.1.0/24 | --- | left |
| 10.0.2.0/24 | --- | right |

# steps at the router



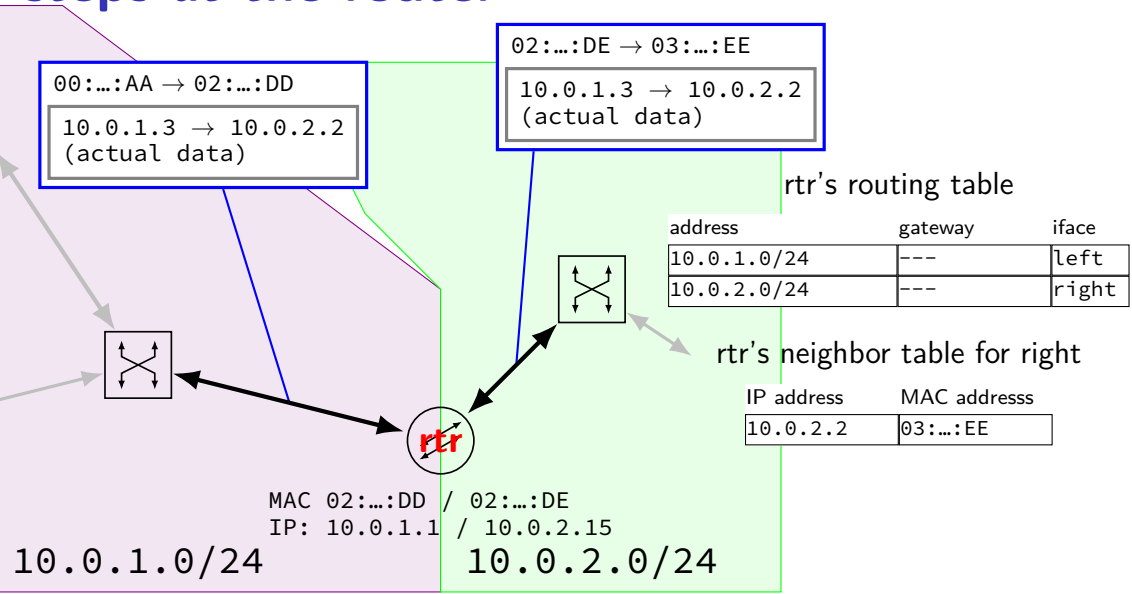

rtr's routing table

| address | gateway | iface |
|---|---|---|
| 10.0.1.0/24 | --- | left |
| 10.0.2.0/24 | --- | right |

rtr's neighbor table for right

| IP address | MAC addresss |
|---|---|
| 10.0.2.2 | 03:…:EE |

# steps at the router




rtr's routing table

| address | gateway | iface |
|---|---|---|
| 10.0.1.0/24 | --- | left |
| 10.0.2.0/24 | --- | right |

rtr's neighbor table for right

| IP address | MAC addresss |
|---|---|
| 10.0.2.2 | 03:…:EE |

# making neighbor tables

need neighbor table to use IP addresses on network

some options:

- system administrator manually configures
- discover dynamically

# manual neighbor tables

on Linux, can run some commands

```
ip niegh add 10.0.2.2 dev right
          lladdr 03:05:…:EE permanent
```

(newer interface, also supports IPv6)

```
arp -i right -s 10.0.2.2 03:05:…:EE
```

IPv4 only; does not allow setting validity duration

# ARP/ND protocols

filling in tables dynamically?

key idea: ask *everyone* on network

entity with that IP address responds

IPv4: Address Resolution Protocol (ARP)

IPv6: ICMPv6 Neighbor Discovery (ND)

ICMP = Internet Control Message Protocol

# ARP messages

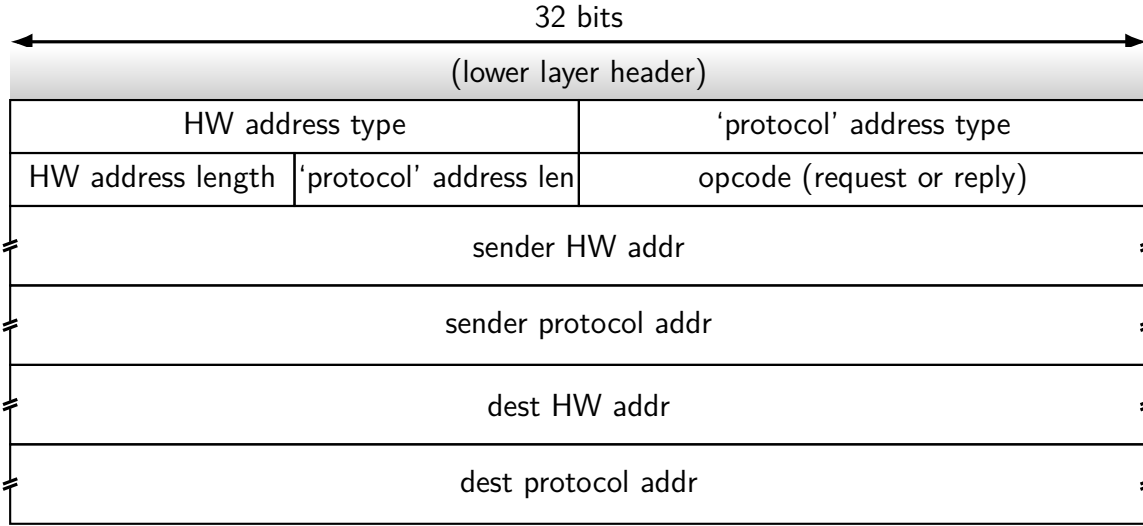
suppose router IP address 10.0.2.15 and MAC address 02:...:DE
needs to find out that 10.0.2.2 uses 03:...:EE

02:...:DE→FF:FF:FF:FF:FF:FF: request 10.0.2.2
tell 10.0.2.15 (=02:...:DE)

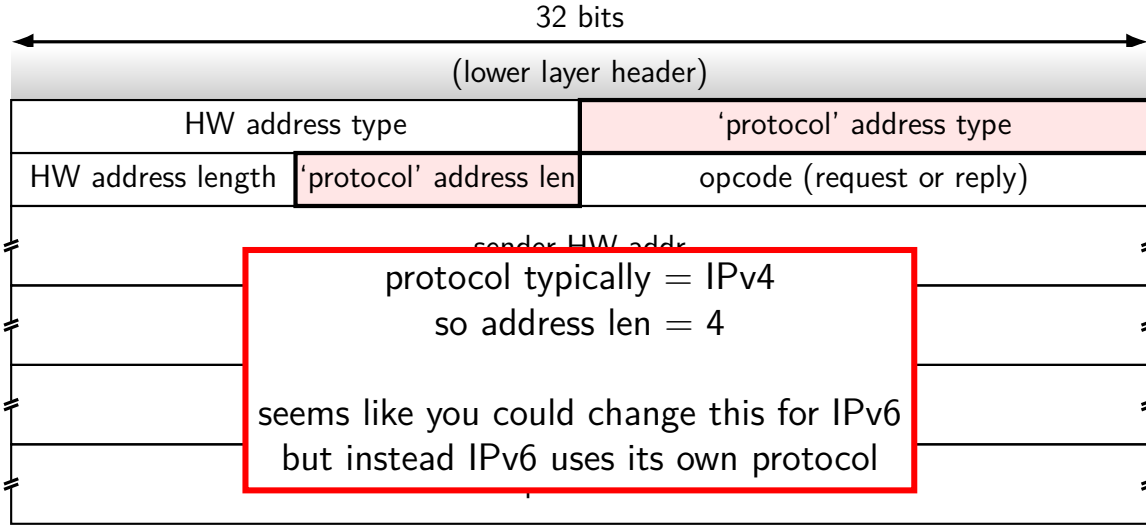
FF:FF:FF:FF:FF:FF = broadcast (send to all on network)

03:...:EE→02:...:DE: reply 10.0.2.2=03:...:EE
tell 10.0.2.15=02:...:DE

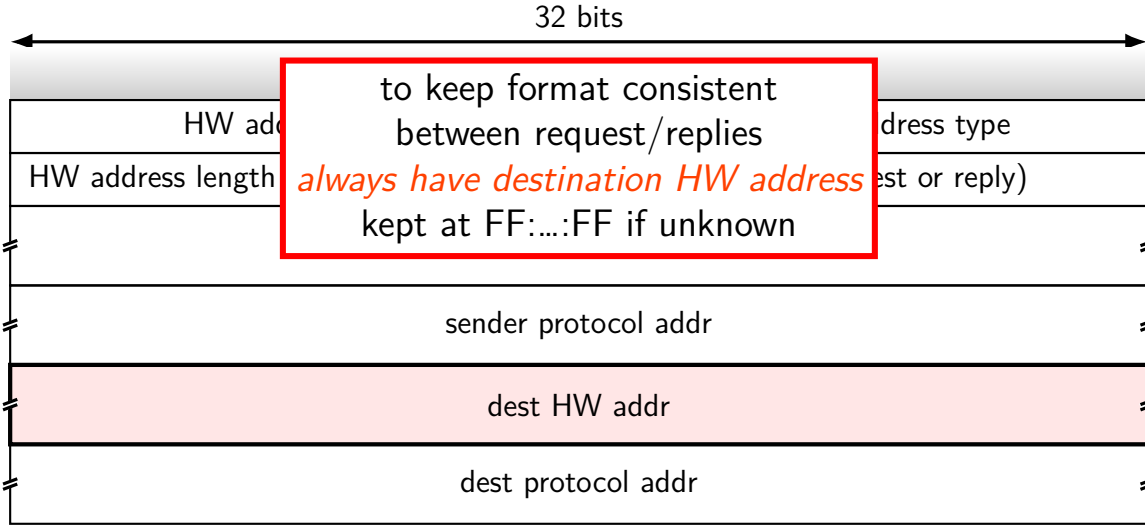
# ARP message format



# ARP message format



# ARP message format



# ARP messages (revisited)

suppose router IP address 10.0.2.15 and MAC address 02:...:DE
needs to find out that 10.0.2.2 uses 03:...:EE

02:...:DE→FF:FF:FF:FF:FF:FF: request 10.0.2.2
tell 10.0.2.15 (=02:...:DE)

*everyone who receives this can add 10.0.2.15=02:...:DE to neighbor table*

03:...:EE→02:...:DE: reply 10.0.2.2=03:...:EE
tell 10.0.2.15=02:...:DE

*everyone who receives this can add 10.0.2.2=03:...:EE to neighbor table*

# ICMPv6 ND

IPv6 uses different protocol for this

...but mostly works the same

differences:

sent as IPv6 packets

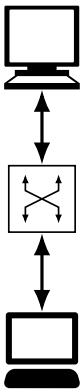
requests sent to special multicast address

- goal: allow nodes to easily ignore irrelevant requests

different names:

- request = solicitiation
- reply = advertisement

MAC 77:…:BB
IP 10.0.2.2

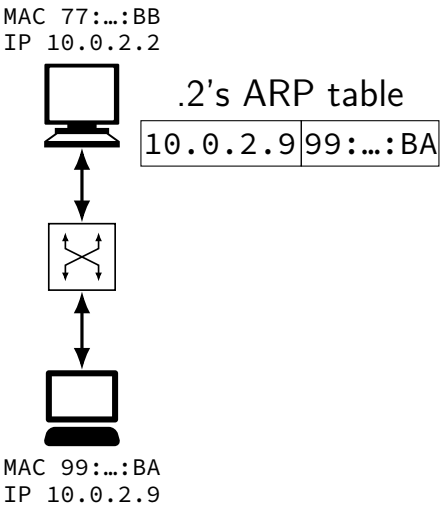


.2's ARP table

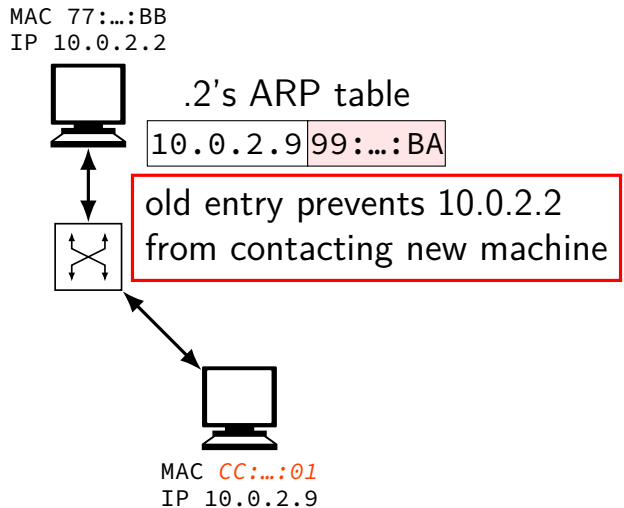
| | |
|---|---|
| 10.0.2.9 | 99:…:BA |

MAC 99:…:BA
IP 10.0.2.9

## Monday




## Tuesday



# gratituous ARP requests

solution: send *unsolicited* ARP messages

CC:...:01→FF:...:FF: request: who has 10.0.2.9, tell 10.0.2.9=CC:...:01

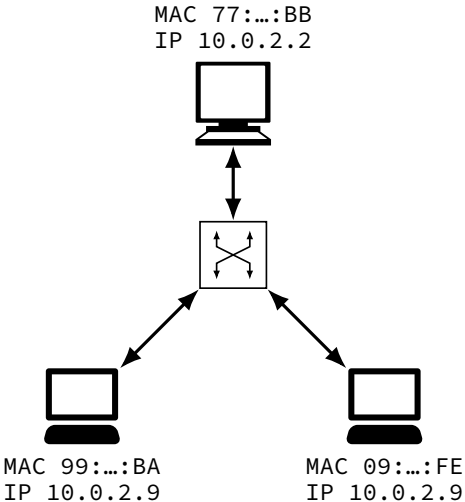
# gratituous ARP requests

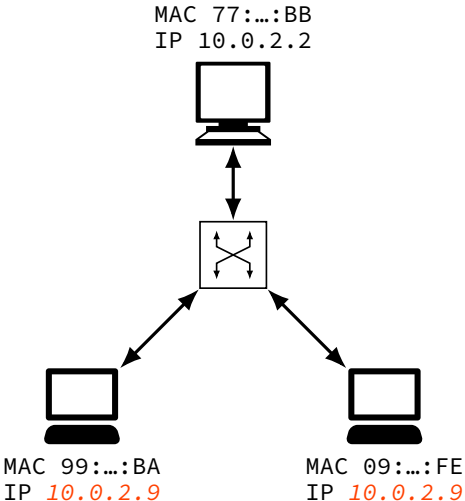
solution: send *unsolicited* ARP messages

CC:...:01→FF:...:FF: request: who has 10.0.2.9, tell 10.0.2.9=CC:...:01

request not reply b/c concerns about old/broken implementations

- ICMPv6 ND fixes this:
  message is 'advertisement' ($\sim$ reply), not 'solicitation' ($\sim$ request)


MAC 77:…:BB
IP 10.0.2.2
MAC 99:…:BA
IP 10.0.2.9
MAC 09:…:FE
IP 10.0.2.9


MAC 77:…:BB
IP 10.0.2.2
MAC 99:…:BA
IP 10.0.2.9
MAC 09:…:FE
IP 10.0.2.9

# duplicate addresses

recommendations in RFC 5227 "IPv4 Address Conflict Detection"

probe for IP address before using it

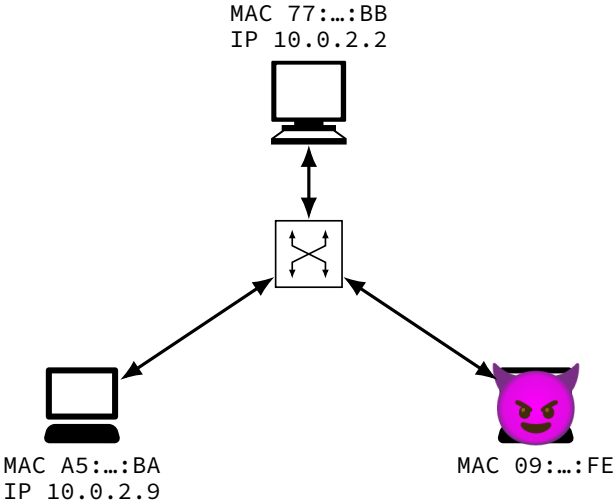
- make sure to broadcast when starting to use address
- probably give up on address if conflict found

watch out for ARP messages indicating address in use

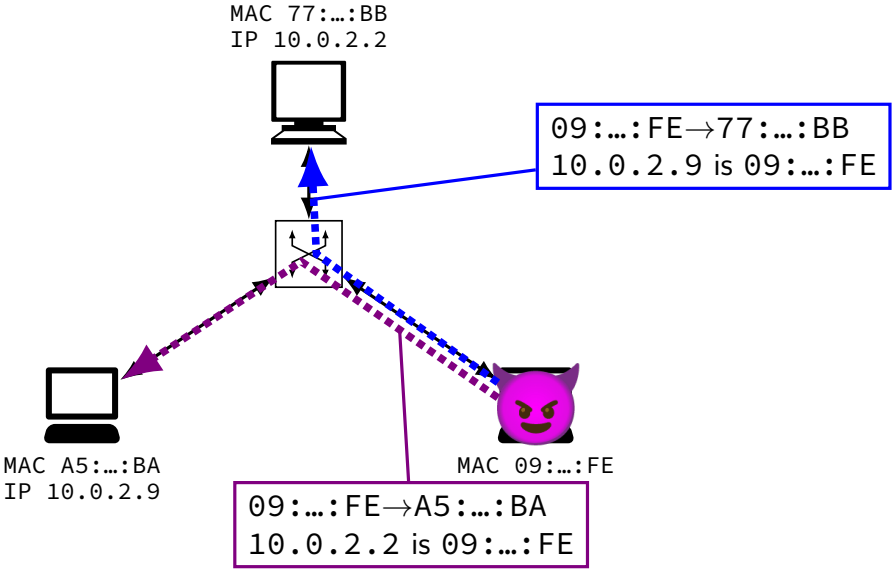
on detecting conflict choose between:

- 'defend' address with more gratituous requests
- give up address

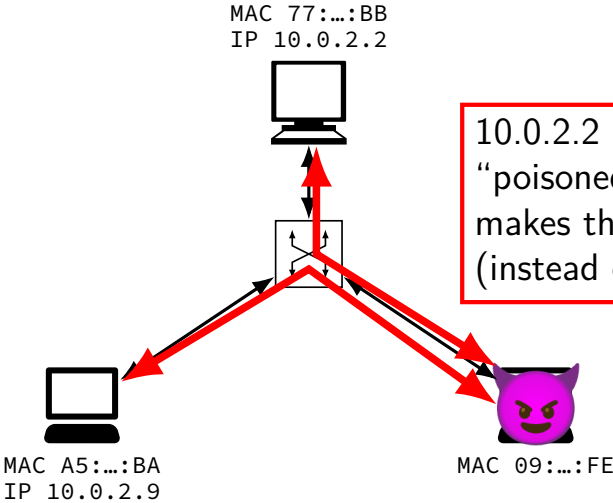
# ARP hijacking



# ARP hijacking

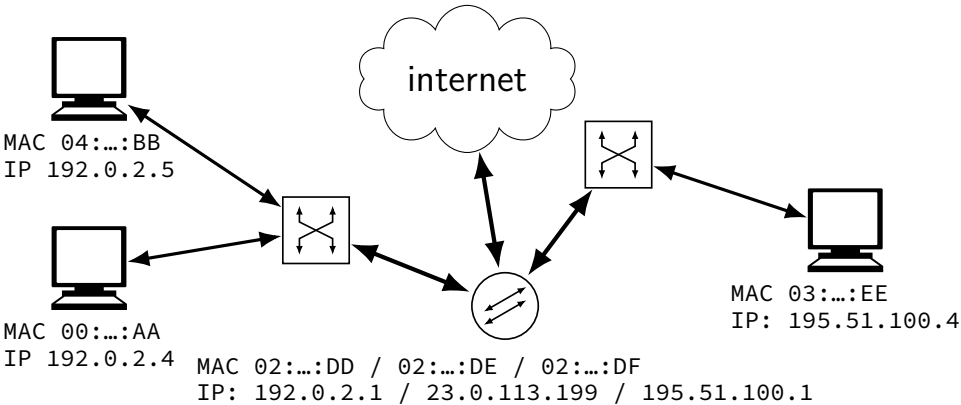


# ARP hijacking



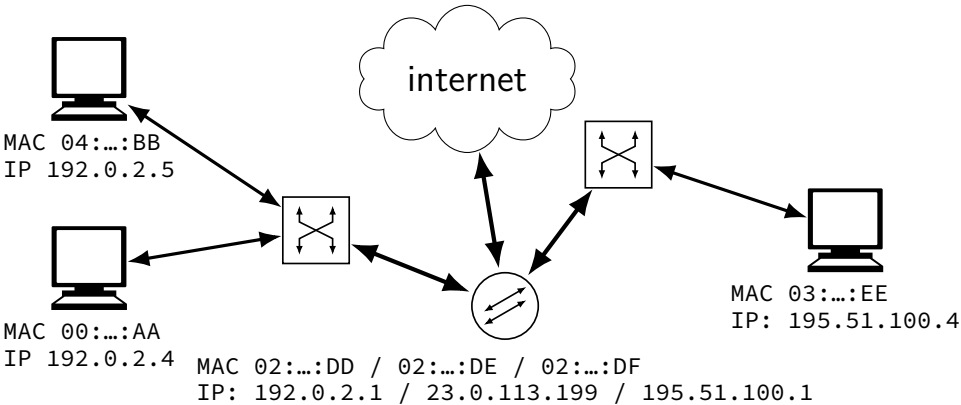

10.0.2.2 and 10.0.2.9 have
"poisoned" ARP tables
makes them send everything to attacker
(instead of each other)

# exercise (1)



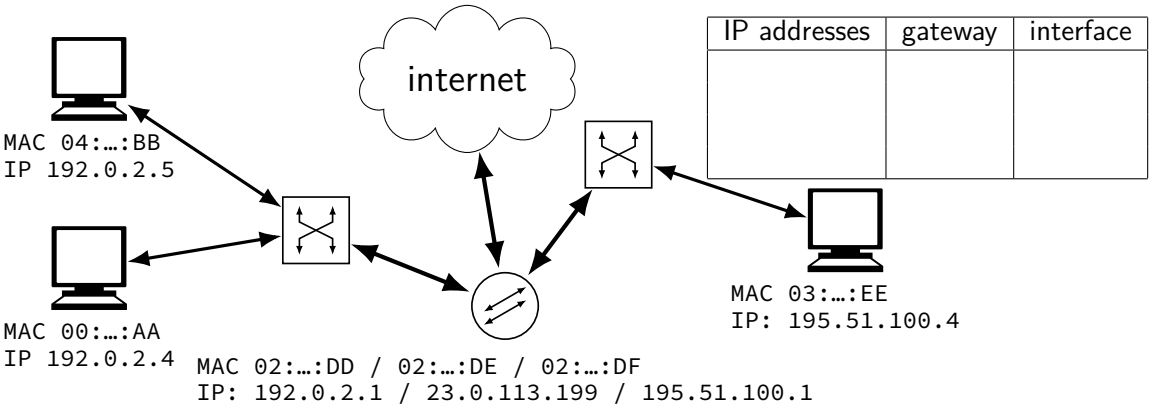

if no one's neighbor tables are setup yet, how many frames get sent when 192.0.2.5 sends one IP packet to 195.51.100.4?

# exercise (2)




what should 192.0.2.5's routing table look like? How about the router's routing table?

# exercise (2)

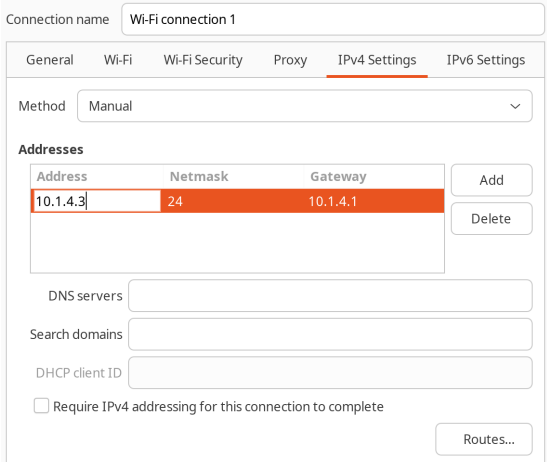



what should 192.0.2.5's routing table look like? How about the router's routing table?

# autoconfiguration

how do hosts get address + default routing table?

one answer: set manually



# simple network config

IP address: 10.0.2.45

(sub)net mask: /25 (aka 255.255.255.128)

varies which format is input

(default) gateway: 10.0.2.102

# simple network config

IP address: 10.0.2.45

(sub)net mask: /25 (aka 255.255.255.128)

varies which format is input

(default) gateway: 10.0.2.102

| addresses | next hop | device |
|---|---|---|
| 10.2.0.0/25 | (direct) | out |
| default | 10.2.0.102 | out |

# DHCP messages (1)

protocol looks weird in packet traces because of history

built on top of UDP + IP

built as extension to older BOOTP (bootstrap protocol)

common message format for different “operations”

# DHCP messages (2)

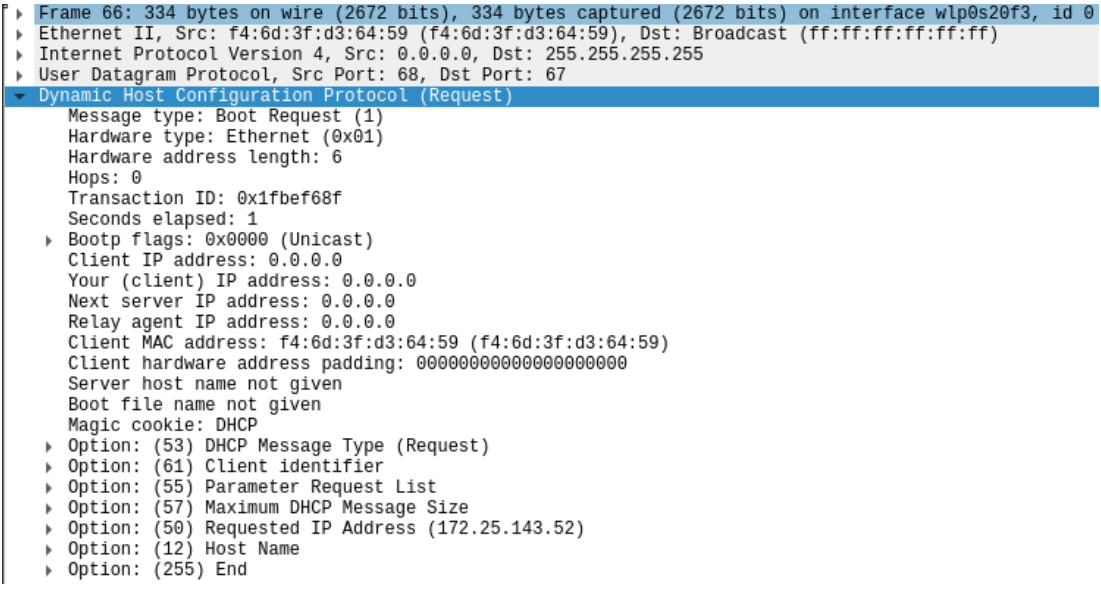
from client (looking to configure itself):

- DISCOVER (look for configuration server)
- REQUEST (get configuration from server)

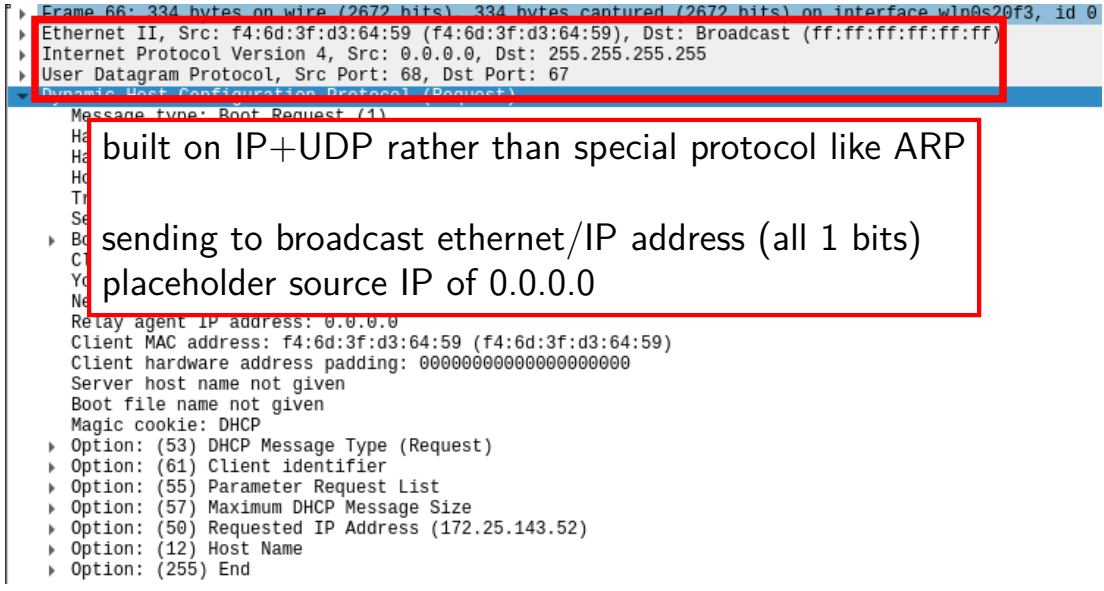
from server (offering configurations):

- OFFER ('I am a configuration server')
- ACK (here's a configuration)

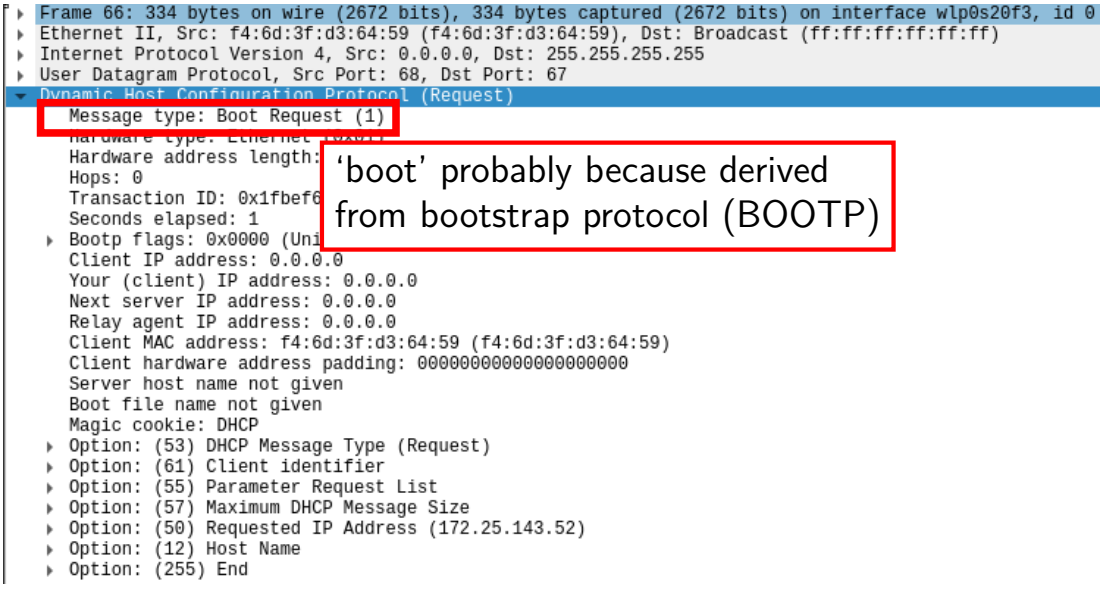
# DHCP request example



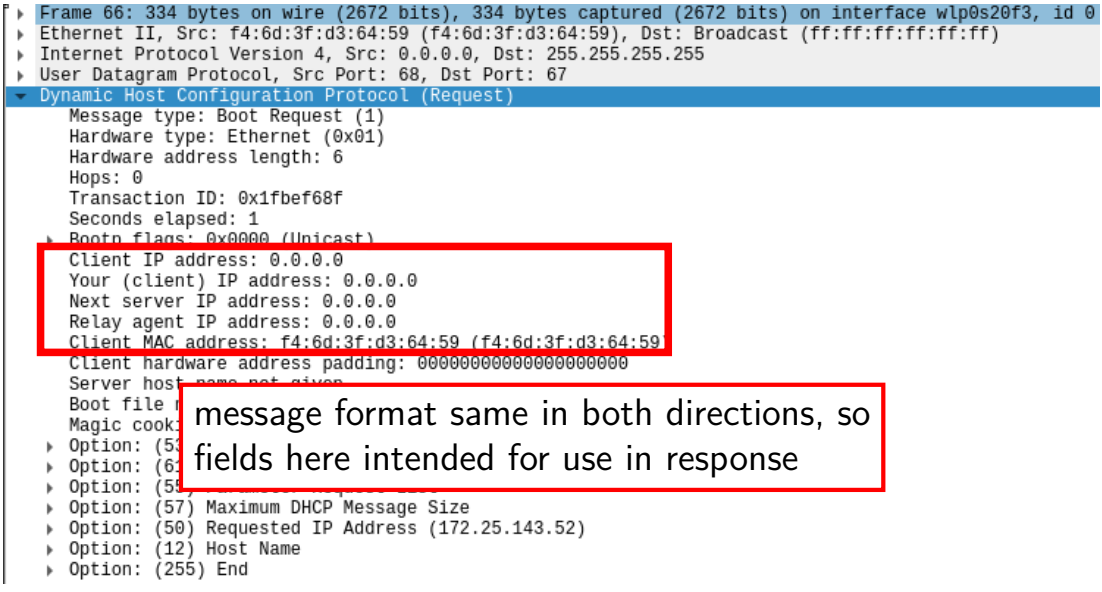
# DHCP request example



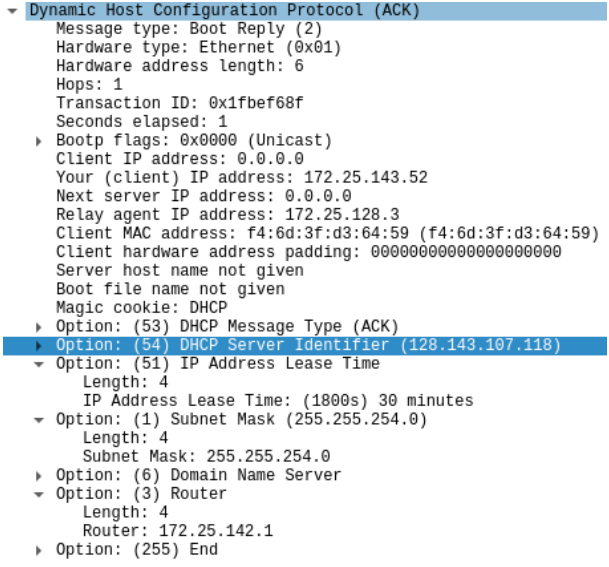
# DHCP request example



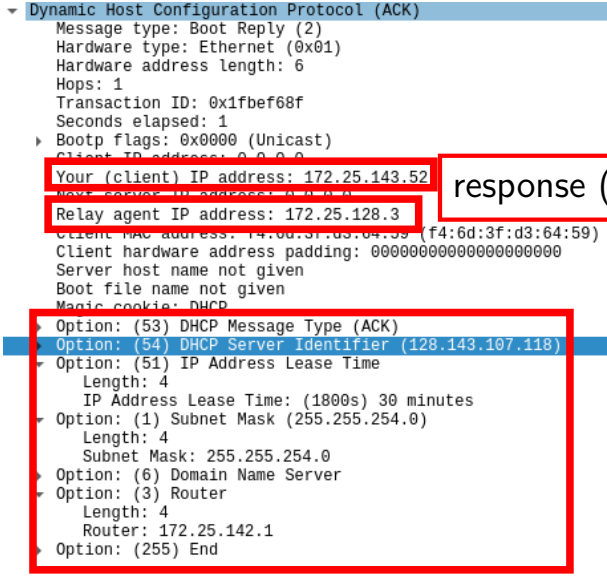
# DHCP request example



# DHCP ACK example

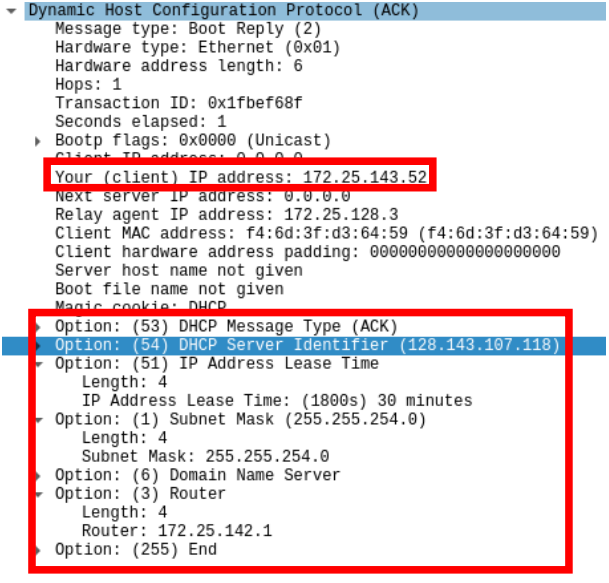


# DHCP ACK example




response (ACK) has address fields filled in

# DHCP ACK example




/23 = 255.255.254.0 m

172.25.142.0 =
172.25.143.52 bitwise-A

# DHCP leases

DHCP ACKs specify a time limit

(example from prior slide (UVa eduroam): 30 minutes)

need to be renewed (new REQUEST + ACK)

REQUESTs can contain 'desired address' (= current address when renewing)

# how many DHCP servers?

DHCP assumption: broadcast to local network and there's the server

conflicting goals:

- want broadcasts not to go too many machines
- want to have few DHCP servers

# how many DHCP servers?

DHCP assumption: broadcast to local network and there's the server

conflicting goals:

- want broadcasts not to go too many machines
- want to have few DHCP servers

solution: DHCP *relays*

# DHCP relays

# IPv6 autoconfiguration

in IPv4, autoconfiguration "bolted on"

- one protocol to be assigned address (DHCP)
- one protoocl to communicate IP address to other nodes on network (ARP)

IPv6 was designed later, so they thought about it early

# big network address assignment

IPv6 local networks are typically /64s

- $2^{64}$ address available for local network

why so big? allow easy address assignment

StateLess Address Auto Configuration (SLAAC)

# MAC-address based address assignment

let's say my local network is 2001:db8:4999:3333::/64

| MAC address | IPv6 address |
|---|---|
| `11:22:33:44:55:66` | `2001:db8:4999:3333:1122:33ff:fe44:5566` |
| `01:A0:B3:CC:DD:FF` | `2001:db8:4999:3333:01a0:b3ff:fecc:ddff` |
| … | … |

# MAC-address based address assignment

let's say my local network is 2001:db8:4999:3333::/64

| MAC address | IPv6 address |
|---|---|
| *11:22:33:44:55:66* | 2001:db8:4999:3333:*1122:33*ff:fe*44:5566* |
| *01:A0:B3:CC:DD:FF* | 2001:db8:4999:3333:*01a0:b3*ff:fe*cc:ddff* |
| … | … |

Network Working Group
Request for Comments: 3041
Category: Standards Track

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IBM
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...
Abstract

..... Use of the extension causes nodes to generate global-scope addresses from interface identifiers that change over time, even in cases where the interface contains an embedded IEEE identifier. Changing the interface identifier (and the global-scope addresses generated from it) over time makes it more difficult for eavesdroppers and other information collectors to identify when different addresses used in different transactions actually correspond to the same node.

# late timeline

privacy extensions weren't default until

- MacOS X 10.7 (2011)
- Windows Vista (2007)
- Ubuntu 12.04 (2012)

# SLAAC

uses ICMPv6 (same as for neighbor discovery)

two modes:

- when there's a router (get *global* addresses)
- when there's only a local network (get *link-local* addresses)

# router adverisements

nodes send a ICMPv6 *Router Solicitation* message

receive back a ICMPv6 *Router Advertisement* which can have:

prefix information
- nodes choose address starting with prefix
- check for duplicates using neighbor discovery

DNS information

“managed configuration” flag
- nodes use DHCPv6 to get configuration

“other configuration” flag
- nodes choose address using prefix, but more info avail via DHCPv6

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# backup slides